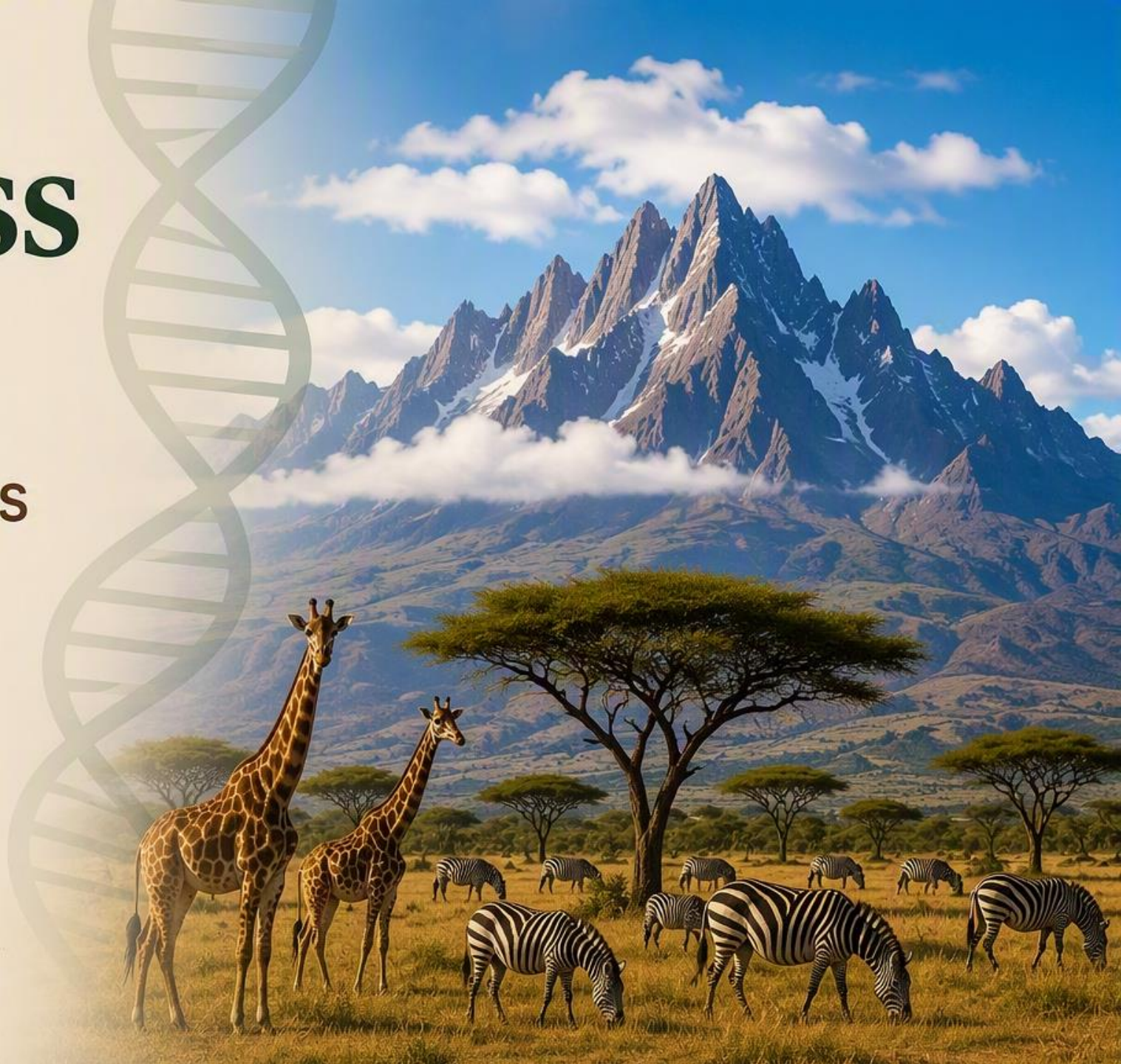
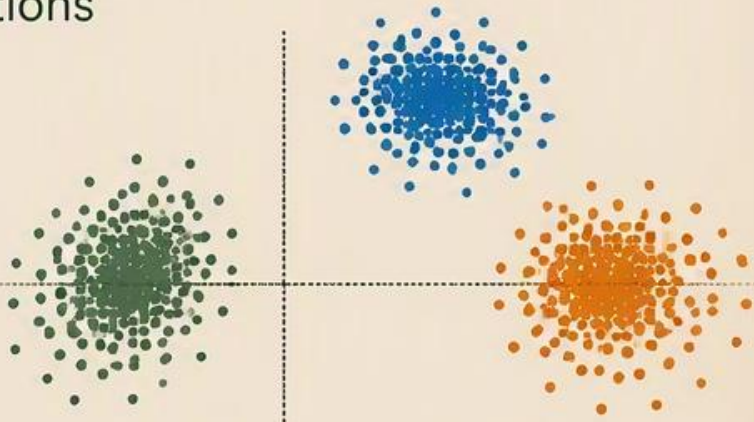


Relatedness and F_{ST}

IN POPULATION GENETICS

Understanding genetic similarity
within and differentiation between
populations



Today's class

Both methods compare observed allele sharing to expected sharing under a demographic model.

Relatedness

How related are two individuals?
IBD sharing vs expectation under
random mating

Relatedness asks: how much more do two individuals share than expected for random individuals?

FST

How differentiated are
populations?
Allele frequency variance among
groups

FST asks: how much do allele frequencies differ among populations compared to within?

FST: Variance in Allele Frequencies

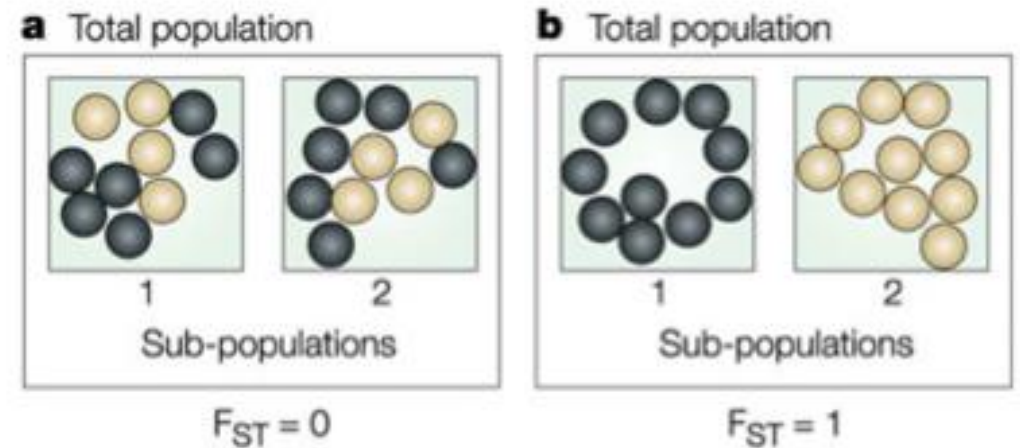
FST – fixation index

How differentiated are
populations?
Allele frequency variance among
groups

**FST measures the proportion of genetic variance
found among populations rather than within them**

Low Fst – No differentiation between the two
populations in this allele

High Fst – Two populations are fixed for different alleles
– Maximum differentiation



FST - Genetic differentiation

Interpretation

FST ≈ 0

Subpopulations are genetically identical

FST ≈ 0.05

Low diff. (e.g., European countries)

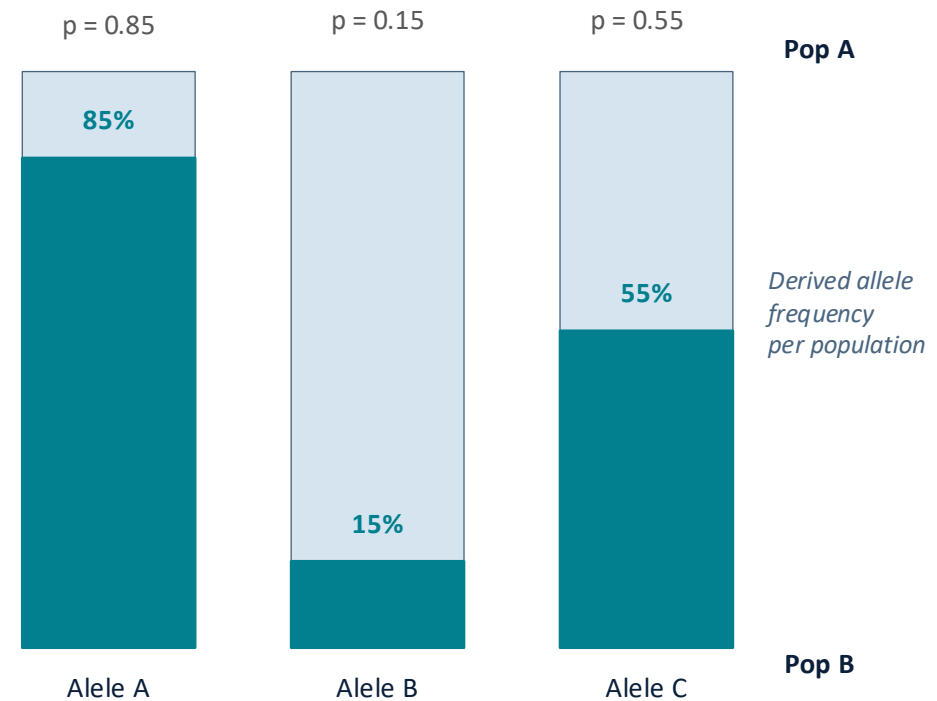
FST ≈ 0.15

Moderate (e.g., human continental groups)

FST ≈ 1

Complete fixation – different alleles fixed

FST $\sim$ proportion of variance between groups



High FST between Pop A & Pop B: allele nearly fixed in A, rare in B.

Weir-Cockerham vs Hudson Estimators

Weir-Cockerham (1984)

Method-of-moments estimator

- Accounts for sample size and number of populations
- Decomposes variance into **within-pop**, **among-pop**, and **total components**
- Handles unequal sample sizes (with caveats)
- Traditional standard for pairwise and global FST
- Implemented in VCFtools, hierfstat, PLINK

Wright (1943) / Weir & Cockerham (1984)

$$F_{ST} = (H_T - H_S) / H_T$$

H_T = expected heterozygosity in total population

H_S = mean expected heterozygosity within subpopulations

Hudson (1992)

Ratio of numerator/denominator

- Designed for **pairwise population comparison**
- More stable for **small sample sizes**
- Used in population genetics software (e.g., EIGENSOFT)
- Often preferred for **per-site FST** in selection scans
- Less biased for pairwise comparisons with unequal n

Hudson (1992)

$$F_{ST} = 1 - H_w / H_b$$

H_w = mean differences within-population

H_b = mean differences between-population

What causes FST?

What causes FST?

● = allele A

● = allele a

1 MUTATION

New alleles arise.



- Changes in DNA create new alleles.
- Source of all new genetic variation.
- Usually occurs rarely.

2 GENE FLOW

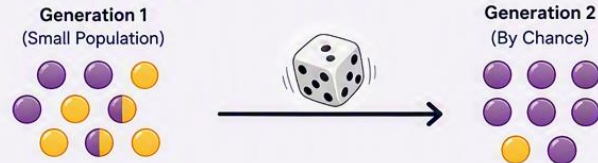
Alleles move between populations.



- Movement of individuals (and their alleles) between populations.
- Introduces new alleles or increases existing ones.
- Increases genetic variation within populations.

3 GENETIC DRIFT

Random changes in allele frequencies.



- Chance events cause alleles to increase or be lost.
- Strongest effect in small populations.
- Can lead to loss of genetic variation.

4 NATURAL SELECTION

Differential survival and reproduction.



- Individuals with advantageous alleles survive and reproduce more.
- Beneficial alleles increase; harmful alleles decrease.
- Leads to adaptation to the environment.



TOGETHER, THESE FORCES SHAPE THE GENETIC MAKEUP OF POPULATIONS OVER TIME.

Mutation creates variation | Gene flow moves variation | Genetic drift shuffles variation | Natural selection favors variation

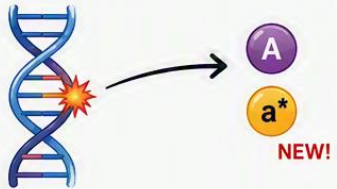
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● = allele A

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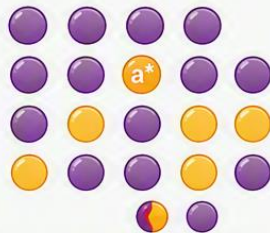
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Population



2 GENE FLOW

Alleles move between populations.



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3 GENETIC DRIFT

Random changes in allele frequencies.

Generation 1
(Small Population)



Generation 2
(By Chance)



- Chance events cause alleles to increase or be lost.
- Strongest effect in small populations.
- Can lead to loss of genetic variation.

4 NATURAL SELECTION

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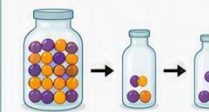
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5 DEMOGRAPHIC HISTORY AND POPULATION STRUCTURE

Changes in population size or structure affect allele frequencies.

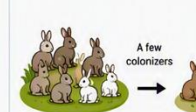
Population Bottlenecks



A drastic reduction in population size by chance reduces genetic variation. When the population grows again, allele frequencies may be very different from the original population.

- ✓ Key idea: Chance events during bottlenecks can strongly change allele frequencies.

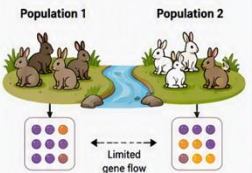
Founder Effects



A new population is started by a small number of individuals. Their allele frequencies may differ from the original population, especially by chance.

- ✓ Key idea: Chance sampling of a few individuals leads to different allele frequencies.

Population Structure
(Subdivision & Limited Gene Flow)



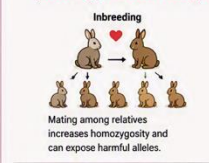
When populations are separated and have limited migration, they accumulate genetic differences over time.

- ✓ Key idea: Reduced gene flow allows populations to evolve differently.

6 NON-RANDOM MATING AND SOCIAL STRUCTURE

Mating patterns and social organization affect allele frequencies.

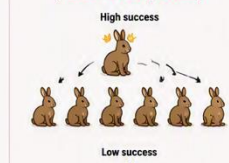
Non-Random Mating
(Inbreeding / Assortative Mating)



Similar mates Dissimilar mates
Individuals choose mates with similar (or dissimilar) traits, changing the distribution of alleles.

- ✓ Key idea: Who mates with whom affects genotype frequencies.

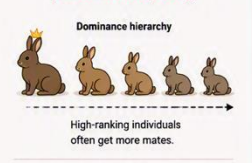
Unequal Reproductive Success
(Variance in Mating Success)



High success Low success
When some individuals leave many more offspring than others, their alleles become more common in the next generation.

- ✓ Key idea: Differences in reproductive success change allele frequencies.

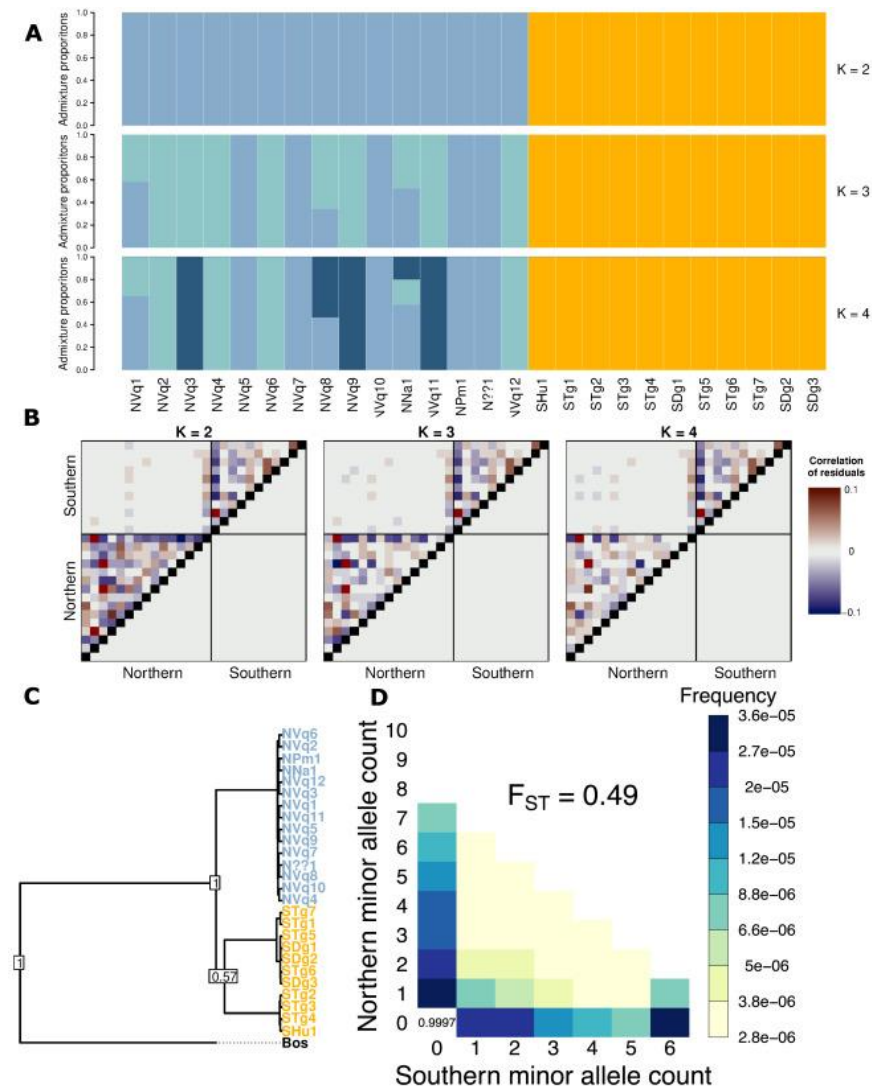
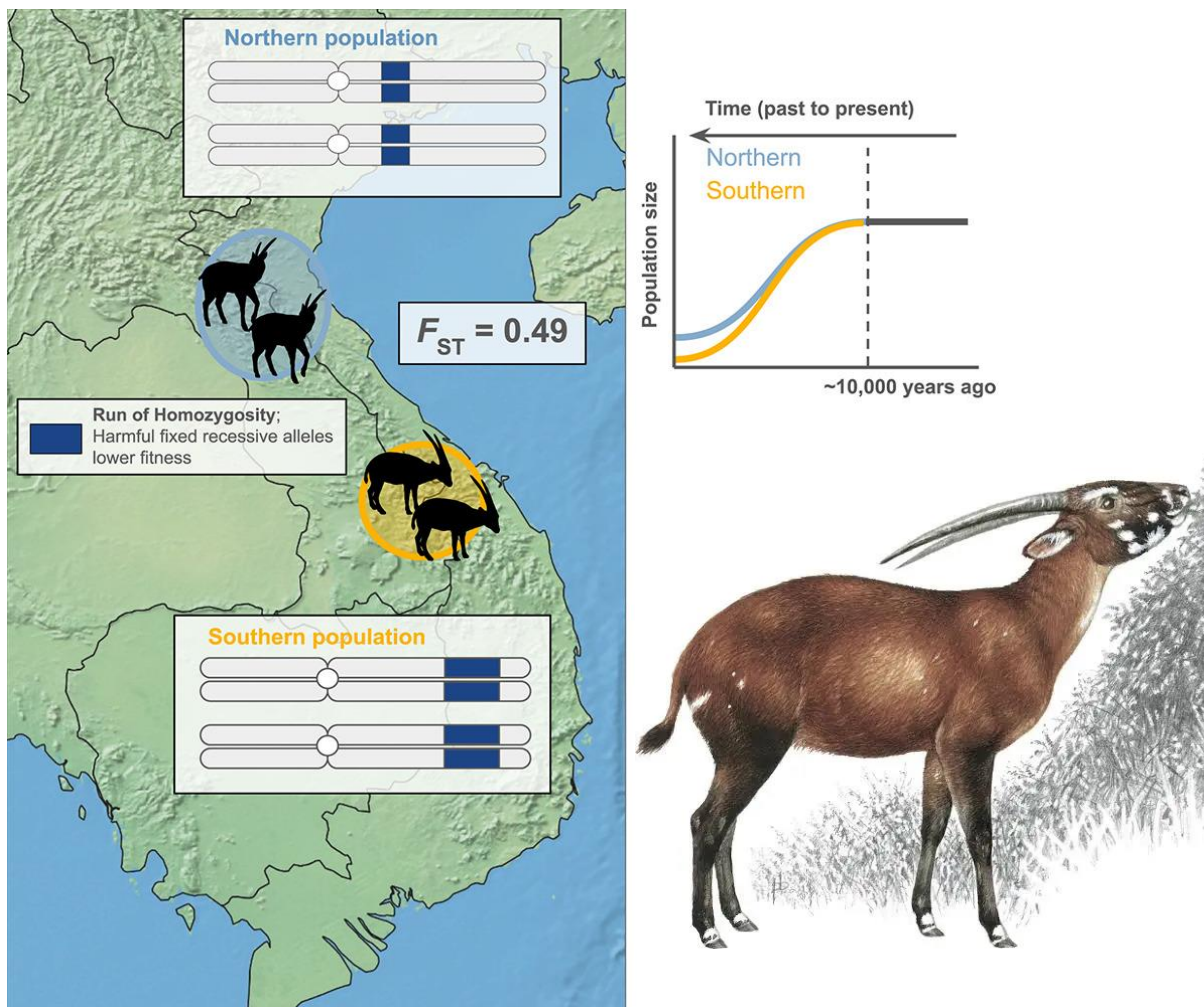
Social Structure
(Hierarchies & Group Living)

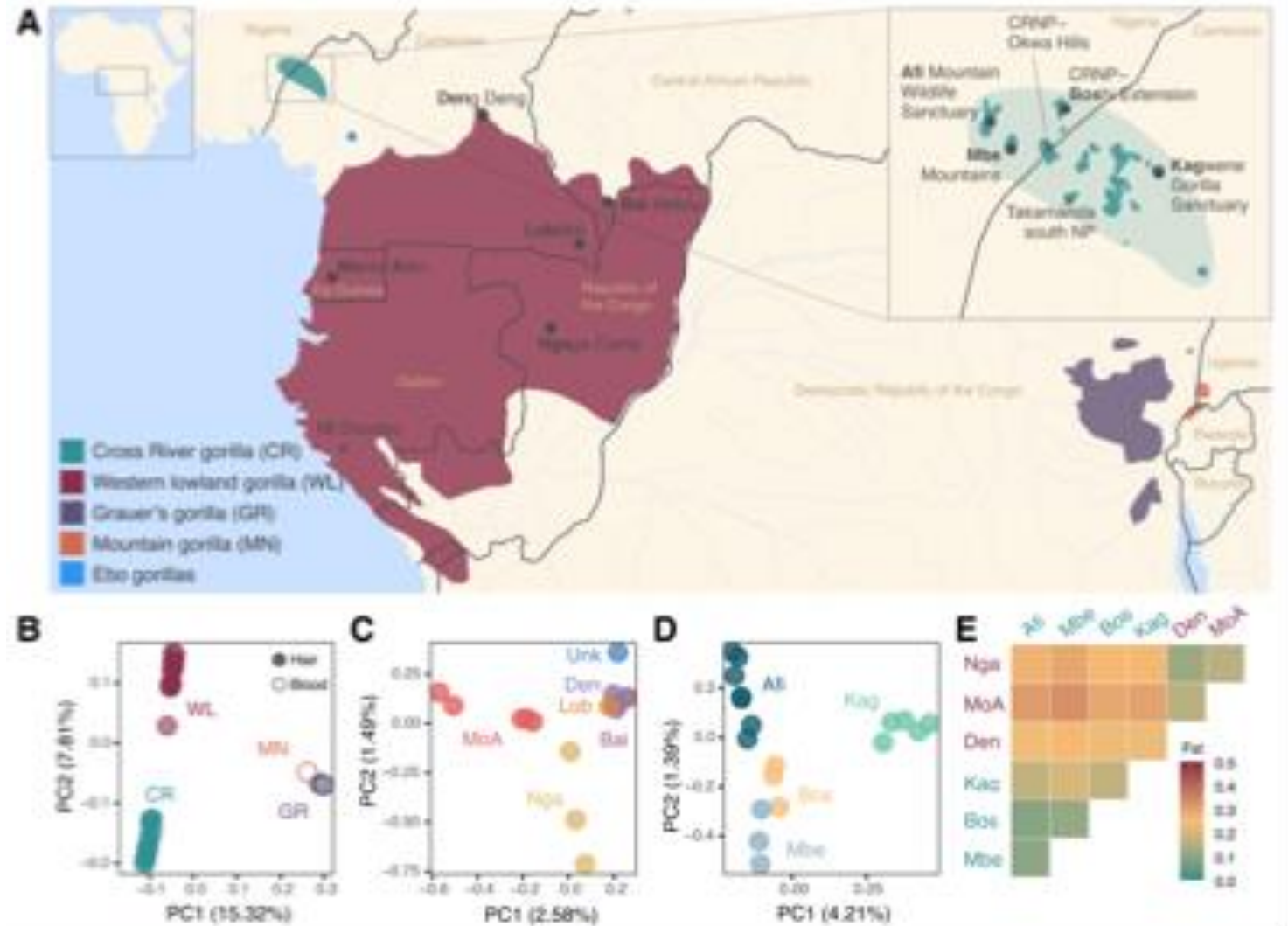
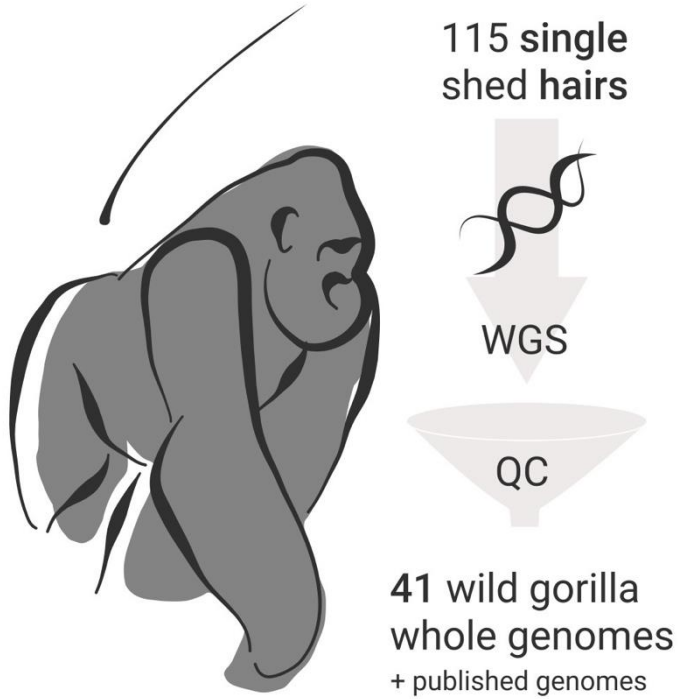


Dominance hierarchy
High-ranking individuals often get more mates.
Group living
Social organization can limit access to mates and resources, affecting which alleles are passed on.

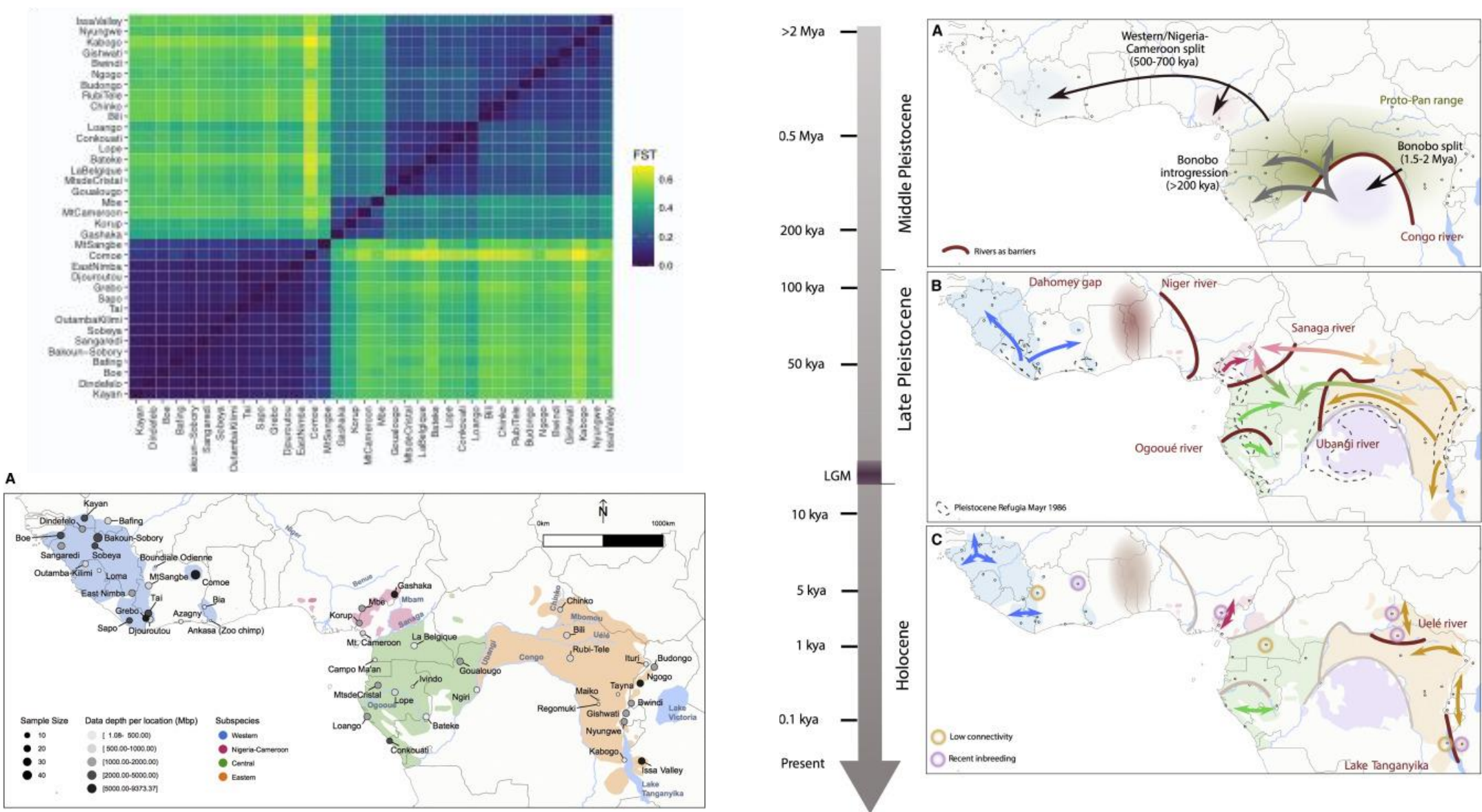
- ✓ Key idea: Social systems influence opportunities to reproduce.

Examples – Saola - fragmentation

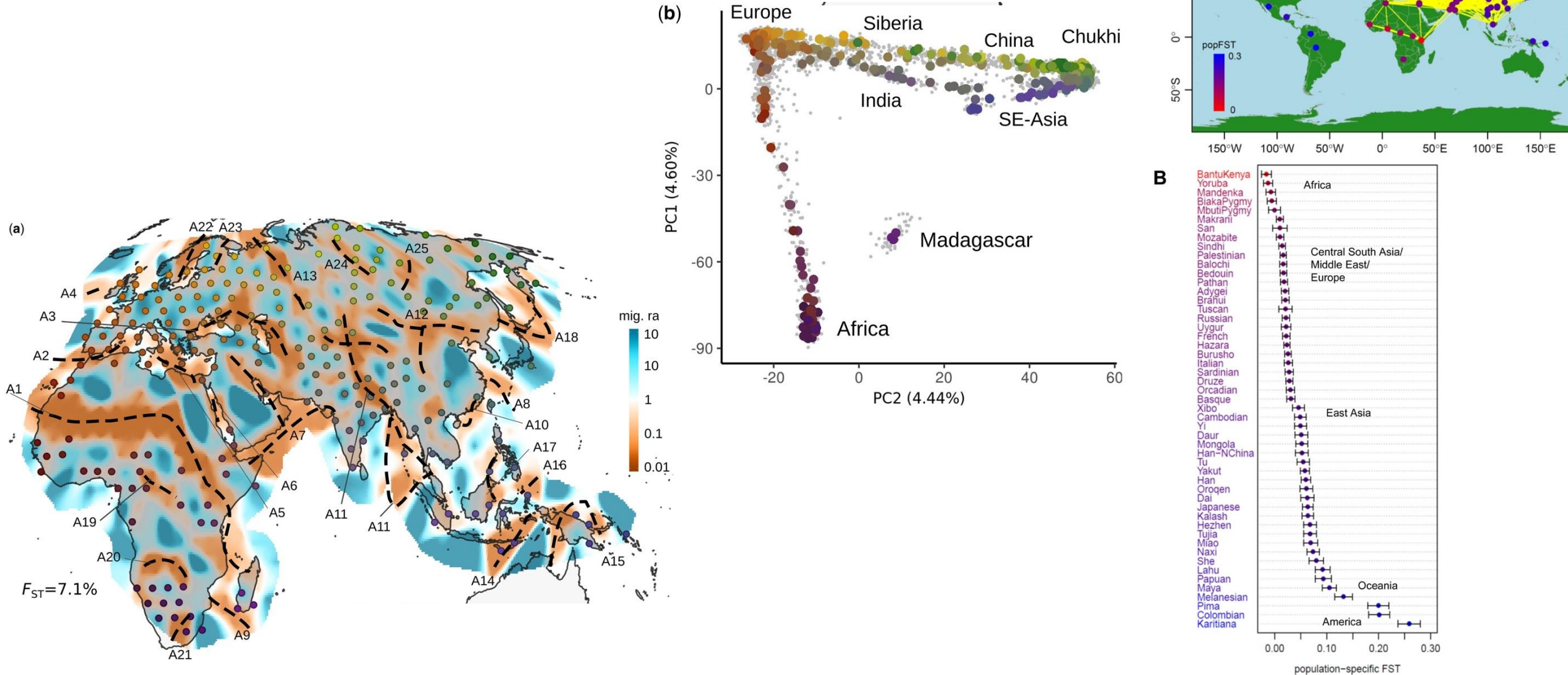




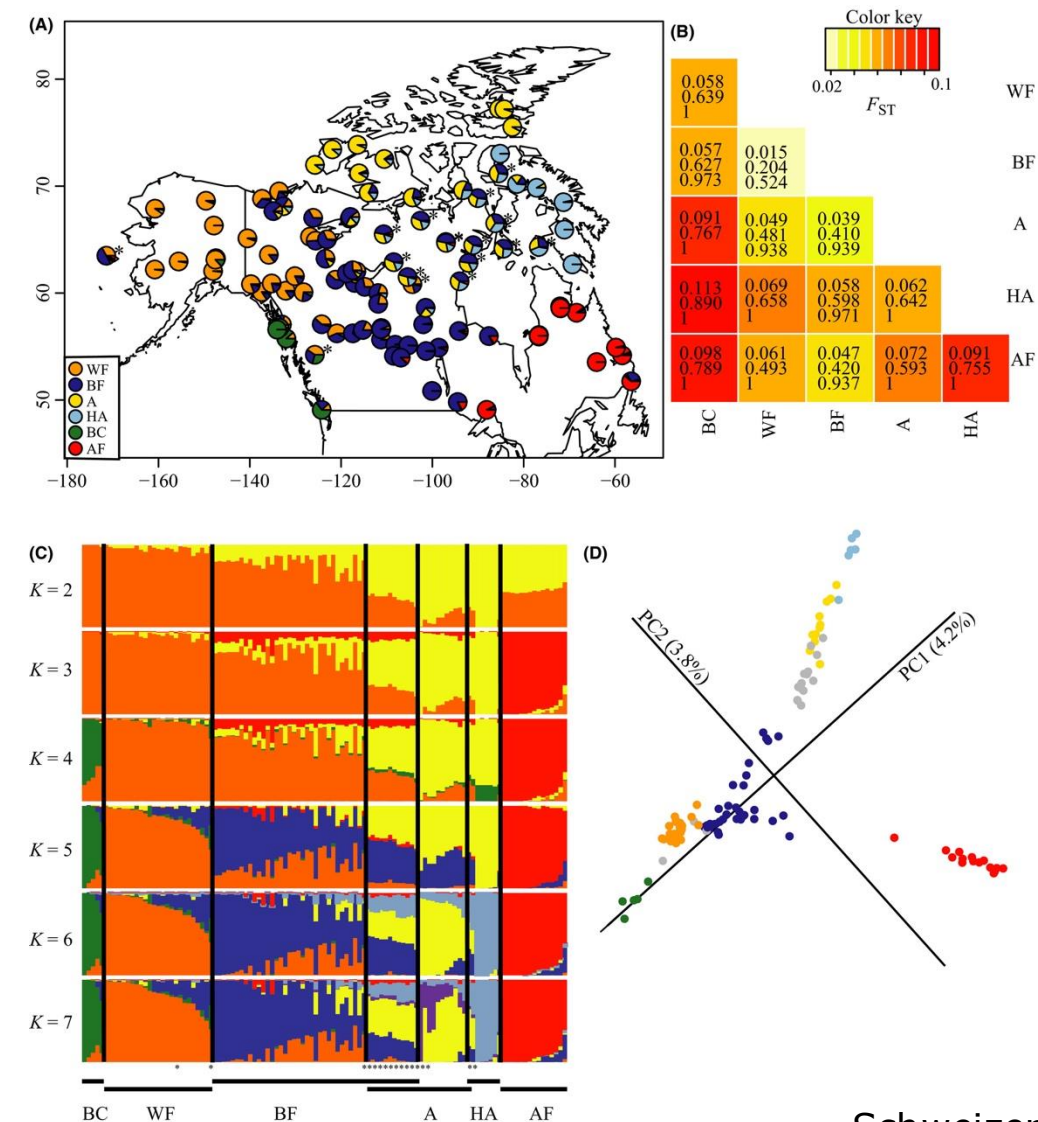
Examples – Chimps - barriers



Examples – Humans – barriers

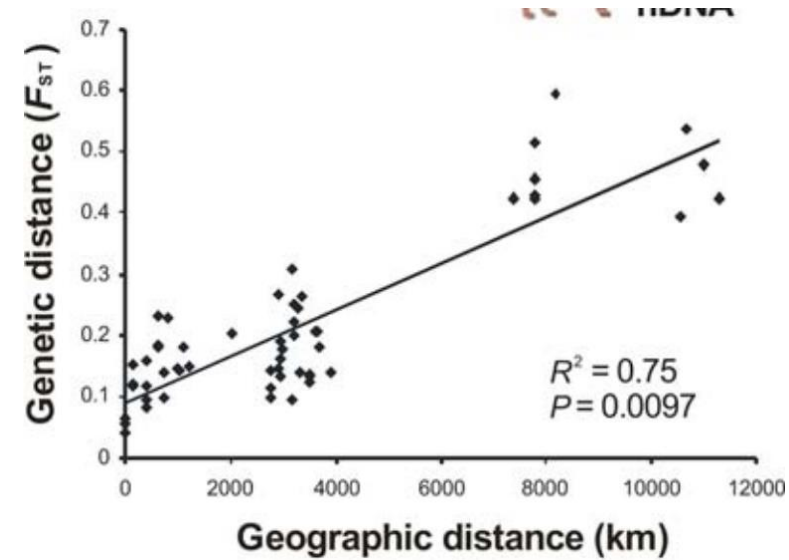
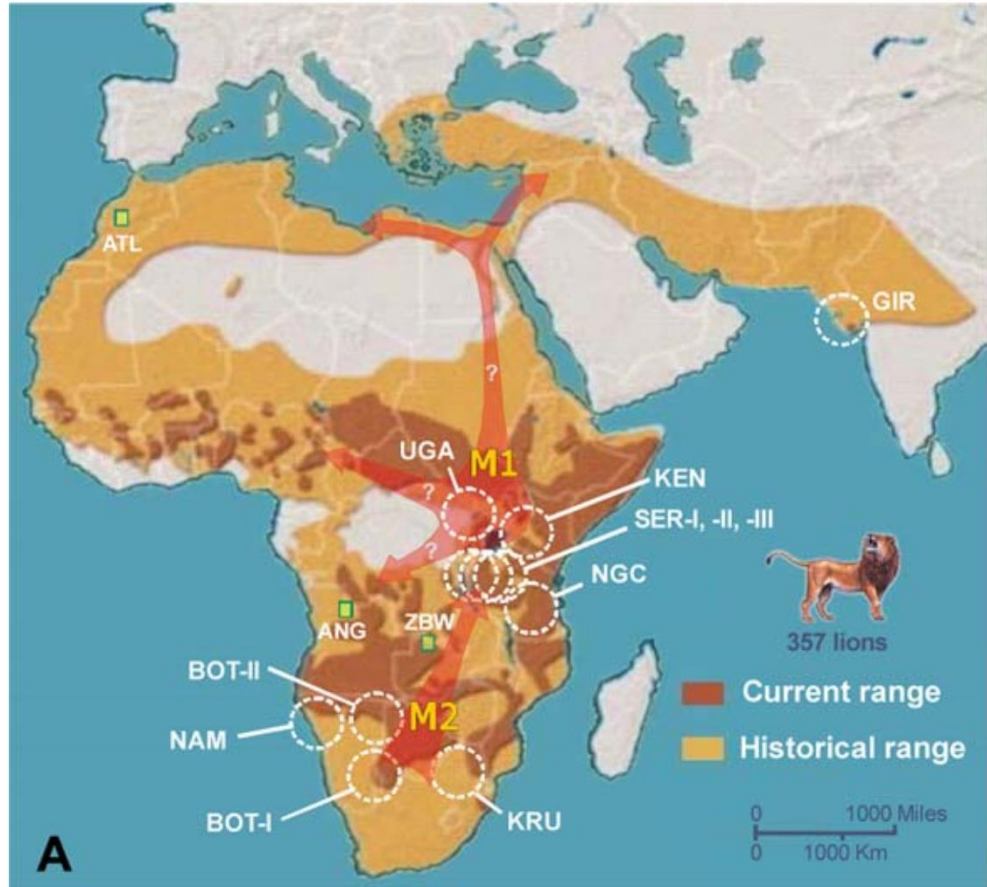


Examples – Wolves – local adaptation



West Forest	Cardiovascular system	Abnormality of the c
	Hearing	Functional abnormal
	Membranes	Integral component
	Metabolism	Metabolic pathways
	Organismal system	Abnormality of the liver
	Skeletal morphology	Abnormality of the external nose
	Vision	Abnormality of the eye
Boreal Forest	Immune response	Immune system processa
	Metabolism	Lipid metabolic processa
	Organismal system	Tissue developmenta
	Response to environment	Response to external stimulusa
	Skeletal morphology	Ossificationa
Arctic	Immune response	Positive regulation of lymphocyte mediated immunity
	Musculature	Abnormality of the musculature
	Organismal system	Functional abnormality of bladder
	Skeletal morphology	Abnormal bone ossification
High Arctic	Brain function	Learning or memory
	Metabolism	Histidine metabolism
British Columbia	Dentition	Misalignment of teeth
	Diet	Salivary secretion
	Metabolism	Arachidonic acid metabolism
	Musculature	Muscle hypertrophy
	Organismal system	Protein transporta
	Skeletal morphology	Disproportionate short stature
Atlantic Forest	Vision	Aplasia/hypoplasia of the iris
	Dentition	Hypodontia
	Metabolism	Glutathione metabolism
	Organismal system	Calcium ion transmembrane transporter activity
	Skeletal morphology	Aplasia involving forearm bones

Examples – Lions – Geographical distance



Scales of FST Estimation

Global

Single FST value across all populations. Useful for overall differentiation. Computed by averaging across loci with variance components. **Comparing species**

Example:
FST = 0.05 across 3 populations

Pairwise

FST between each pair of populations. Reveals which populations are most/least differentiated. **Comparing populations**

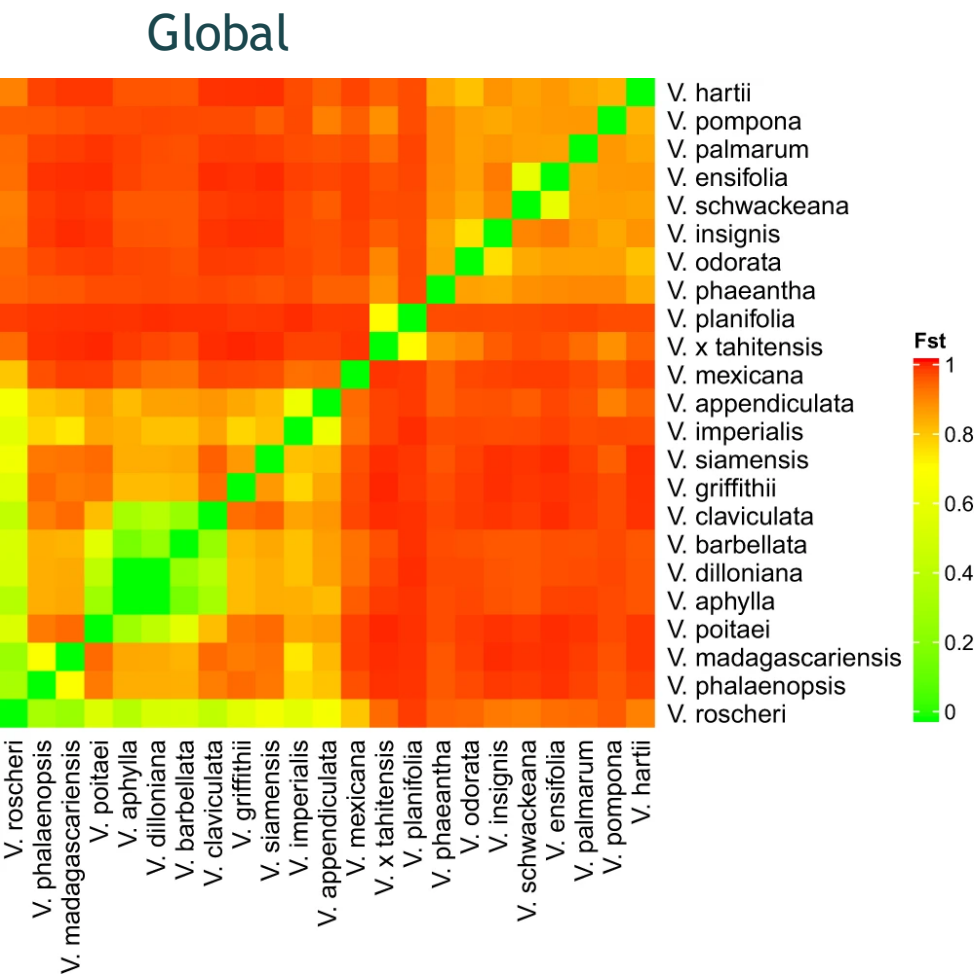
Example:
Pop1-Pop2: 0.03, Pop1-Pop3: 0.12

Per-Site / Windowed

FST at each variant or in sliding windows. Used for outlier scans to detect selection.

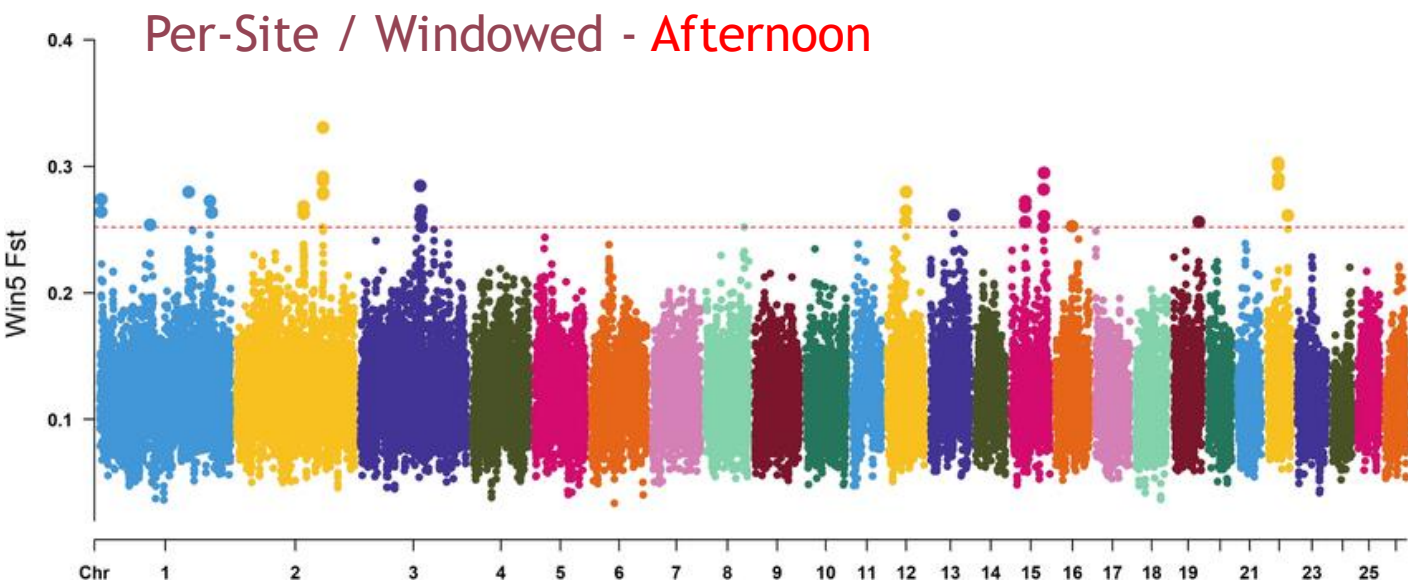
Example:
Window 1Mb: FST = 0.15 (potential outlier)

Scales of FST Estimation



Pairwise
Example of
Giraffes

West African	-	-	-	-	-	-	-	-	-
Kordofan	0.24	-	-	-	-	-	-	-	-
Nubian	0.24	0.16	-	-	-	-	-	-	-
Reticulated	0.35	0.33	0.32	-	-	-	-	-	-
Masai	0.64	0.63	0.62	0.55	-	-	-	-	-
Masai-Selous	0.67	0.66	0.65	0.58	0.10	-	-	-	-
Southern African	0.70	0.69	0.68	0.62	0.63	0.67	-	-	-
Southern Central	0.69	0.68	0.67	0.60	0.62	0.66	0.10	-	-
Angolan	0.71	0.71	0.70	0.63	0.66	0.69	0.19	0.14	-



FST Pitfalls – What can go wrong?

1 Sample size imbalance

Unequal N between populations biases F_{ST} upward for the smaller group.

- Use jackknifing or Weir & Cockerham estimator which partially corrects for this.

2 Ascertainment bias

SNPs discovered in one population are enriched for high-frequency derived alleles in that population, artificially inflating F_{ST} when comparing to others.

- Less relevant with whole-genome sequencing, but still present in some reference panels.

3 Heterozygosity dependency

Its maximum possible value at a locus depends on the frequency of the most common allele and on the number of alleles; for very common or very rare alleles, the attainable F_{ST} can be far below 1

- Low-diversity populations can show inflated F_{ST} even with minor allele frequency differences.

4 Admixture & structure within populations

Cryptic structure within a "population" inflates F_{ST} . Always check PCA first. Relatedness within populations also biases estimates.

- Run PCA first and clean your dataset of related individuals

Time for exercises

Fst in Wildebeest



Black Wildebeest

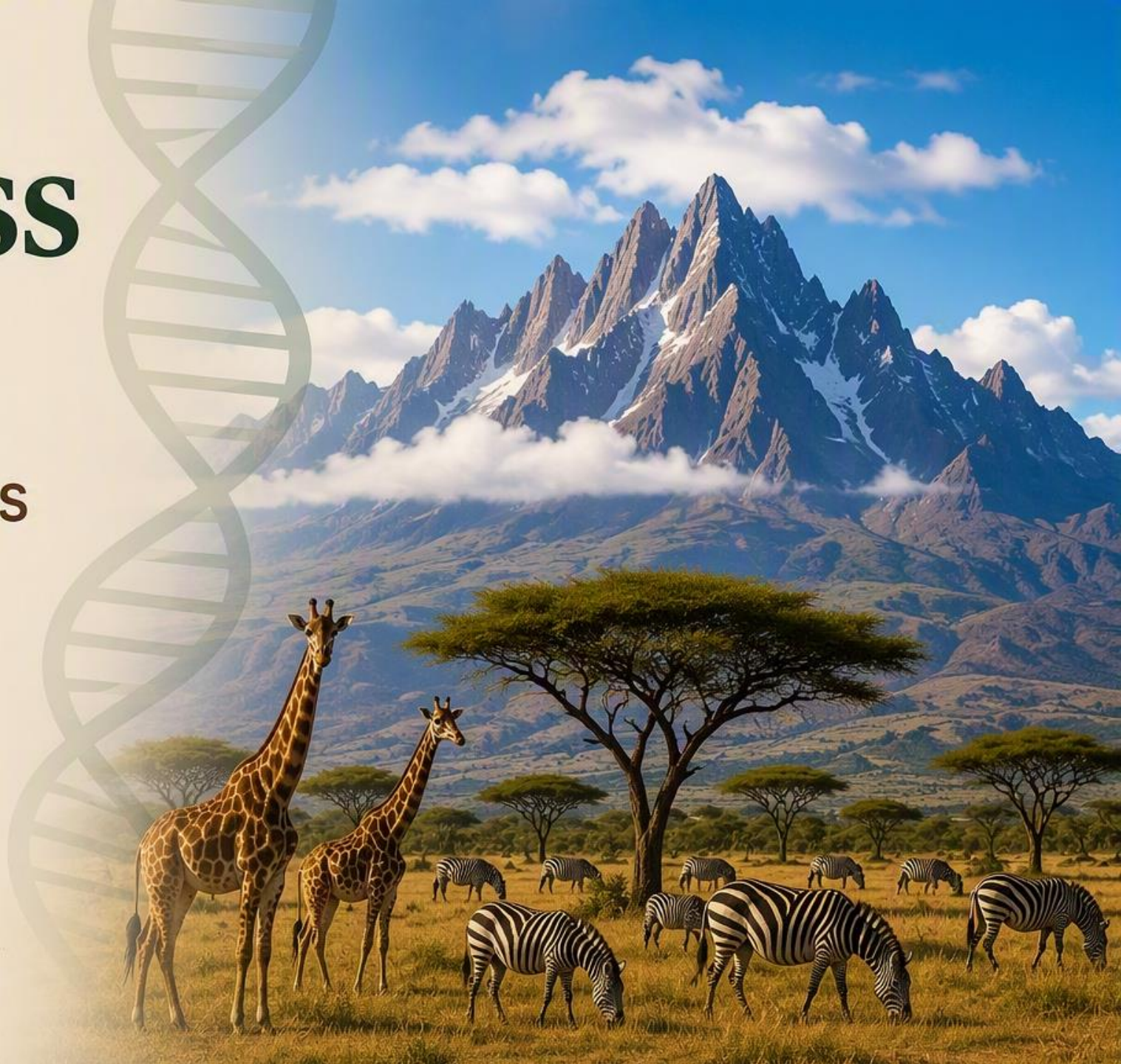
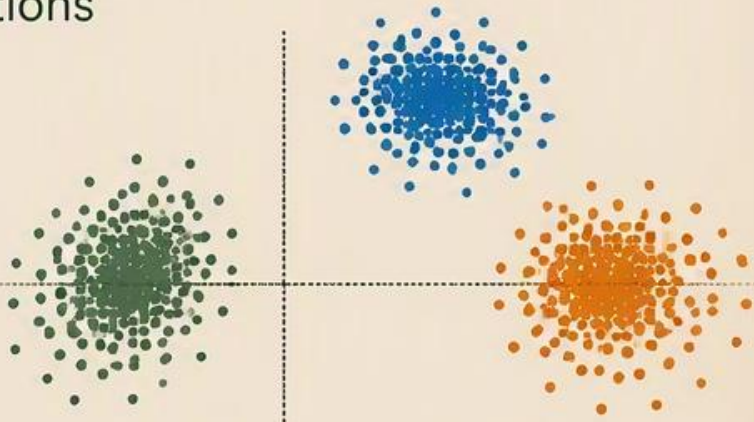


Blue Wildebeest

Relatedness and F_{ST}

IN POPULATION GENETICS

Understanding genetic similarity
within and differentiation between
populations



Today's class

Both methods compare observed allele sharing to expected sharing under a demographic model.

Relatedness

How related are two individuals?
IBD sharing vs expectation under
random mating

Relatedness asks: how much more do two individuals share than expected for random individuals?

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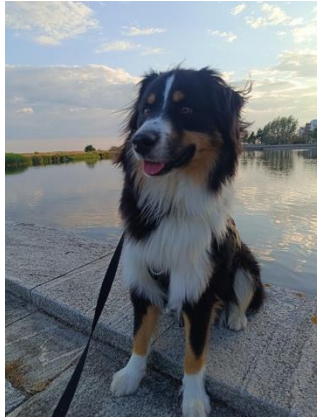
Background:

- What is IBD sharing
- **Inbreeding** and relatedness
- Different relatedness coefficients
- Different methods to measure relatedness

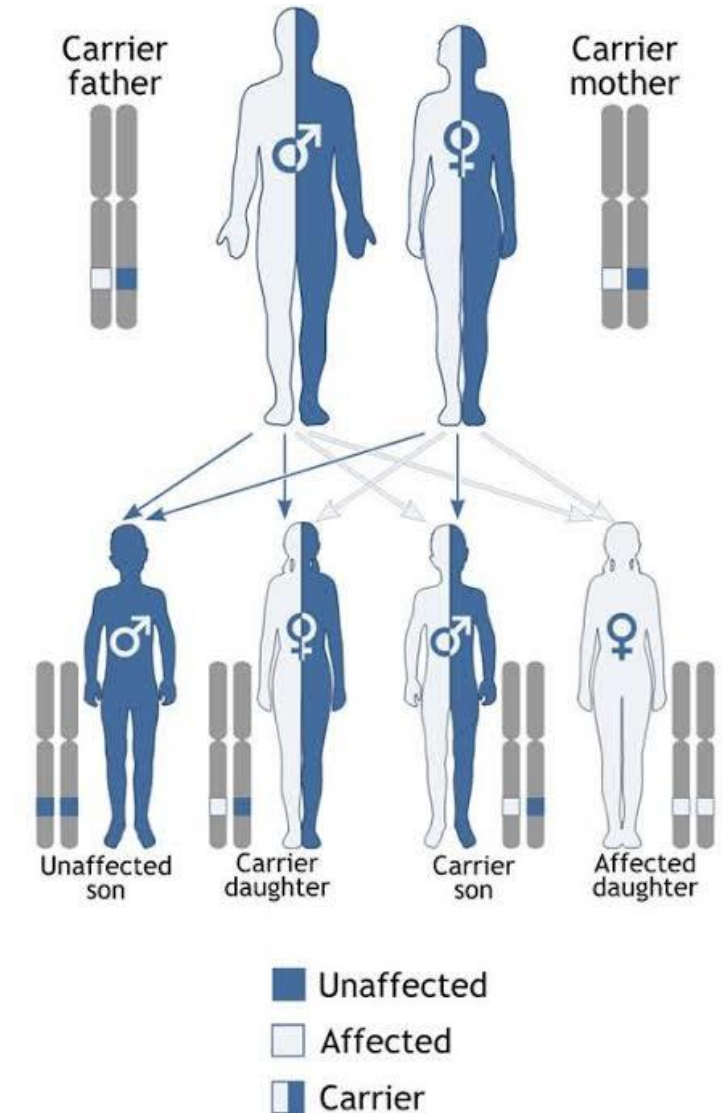
Aims:

- Understand that IBD is not just a number but a pattern across the genome
- How the different coefficients are related
- How relatedness can influence analysis

Relatedness – broad view

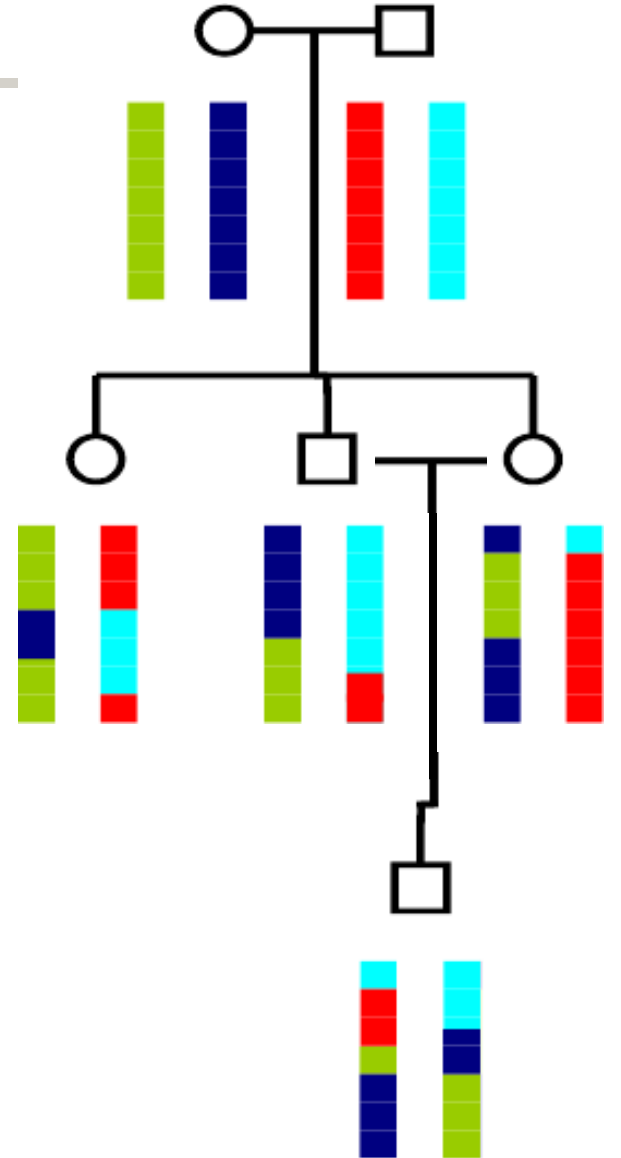


- Meaningful resemblance
- Genes transmission
 - Characteristics
 - Diseases



Not so simple

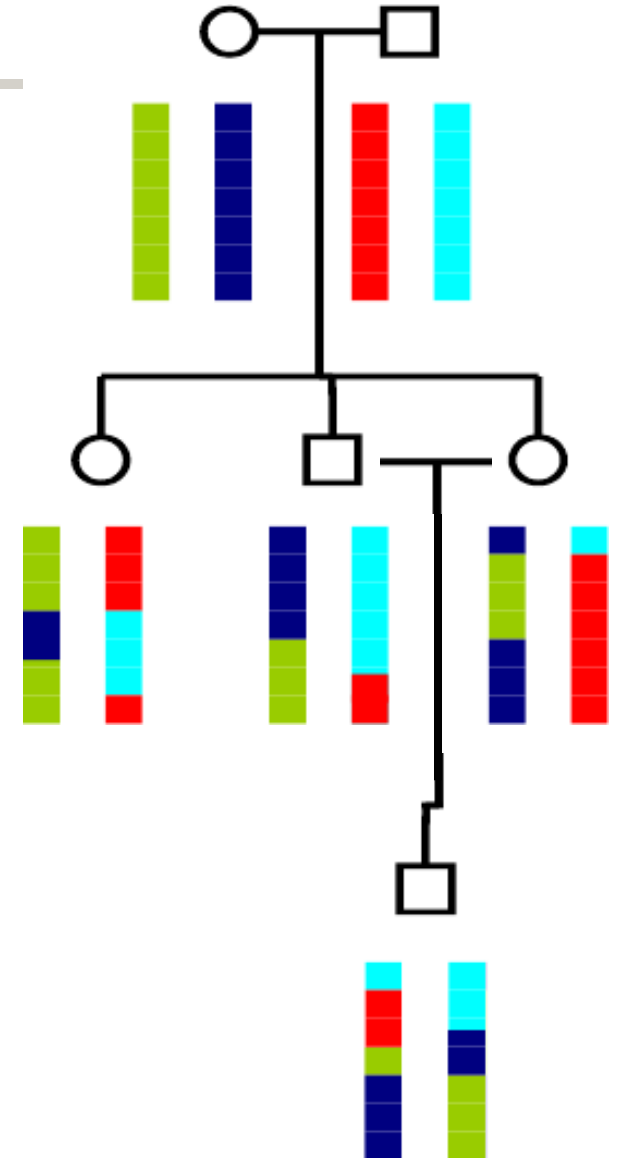
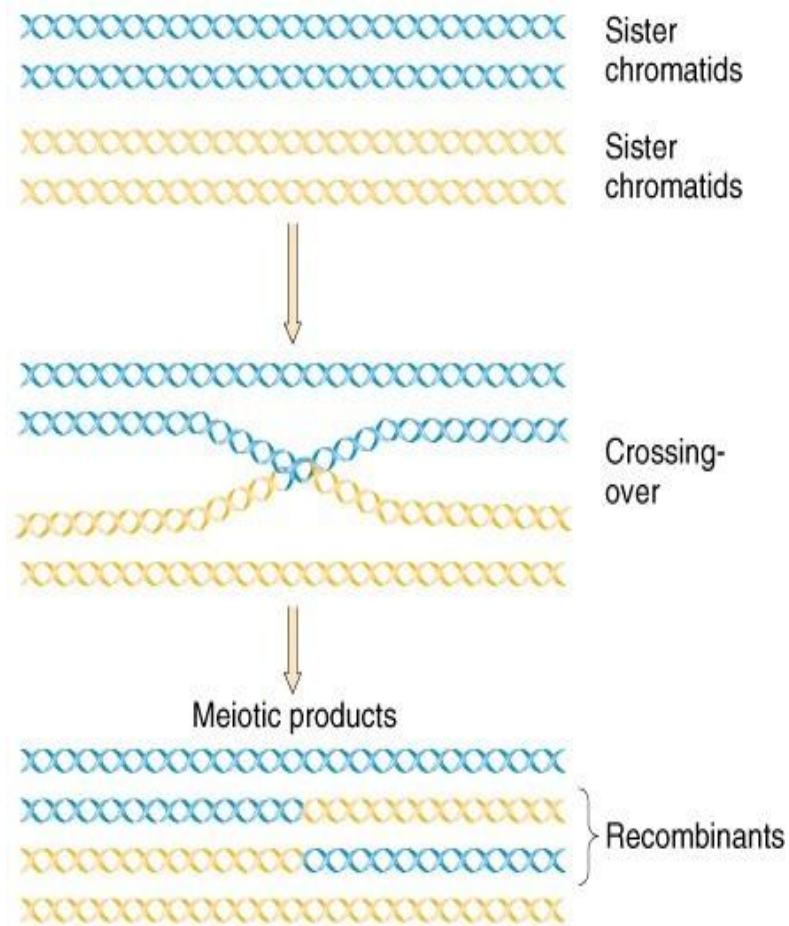
Relatedness between a pair of individuals is usually described using the concept of identity-by-descent (IBD), which is genetic identity due to recent common ancestry.



Not so simple

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Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci



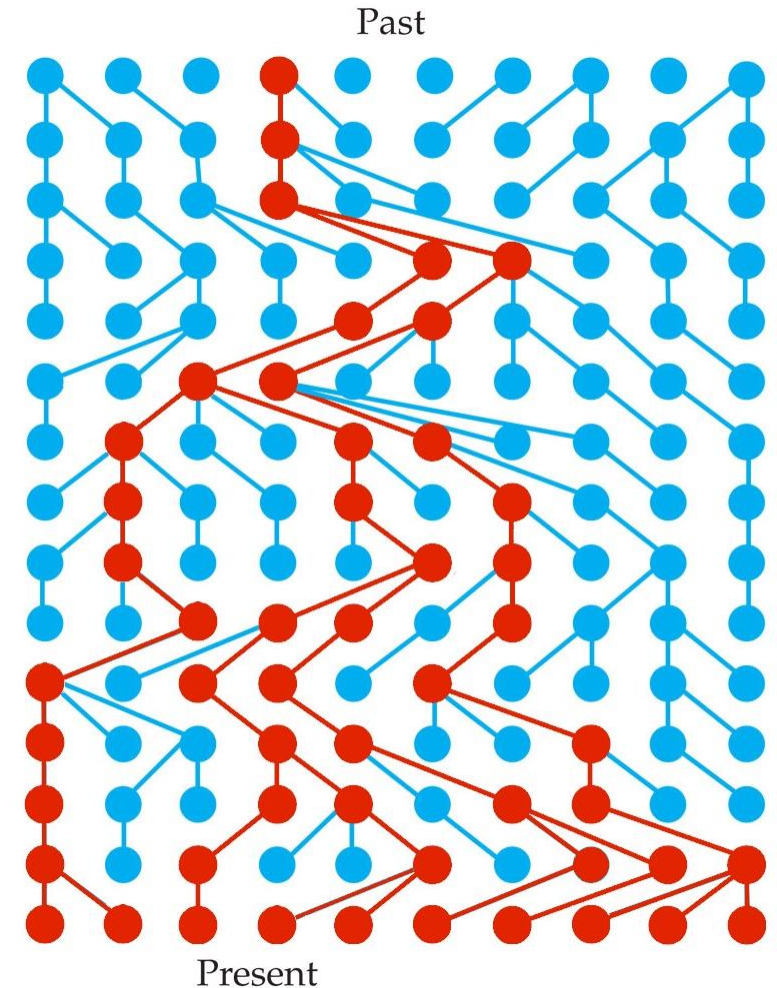
Not so simple

Relatedness between a pair of individuals is usually described using the concept of identity-by-descent (IBD), which is genetic identity due to recent common ancestry.

Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci

All of the individuals in a population are related to each other by a giant pedigree (family tree)

All individuals are related to one another by varying levels of relatedness



IBD vs IBS: Same Genotype, Different Story

Identity by State (IBS)

Two unrelated individuals from different populations can share the same allele (IBS) without being related (IBD).

Individual A



Individual B



*Same alleles (A1, A2) but
different ancestral copies*

Identity by Descent (IBD)

Two alleles are IBD if they are copies of the **same ancestral allele** transmitted through pedigree pathways.

Individual A



Individual B

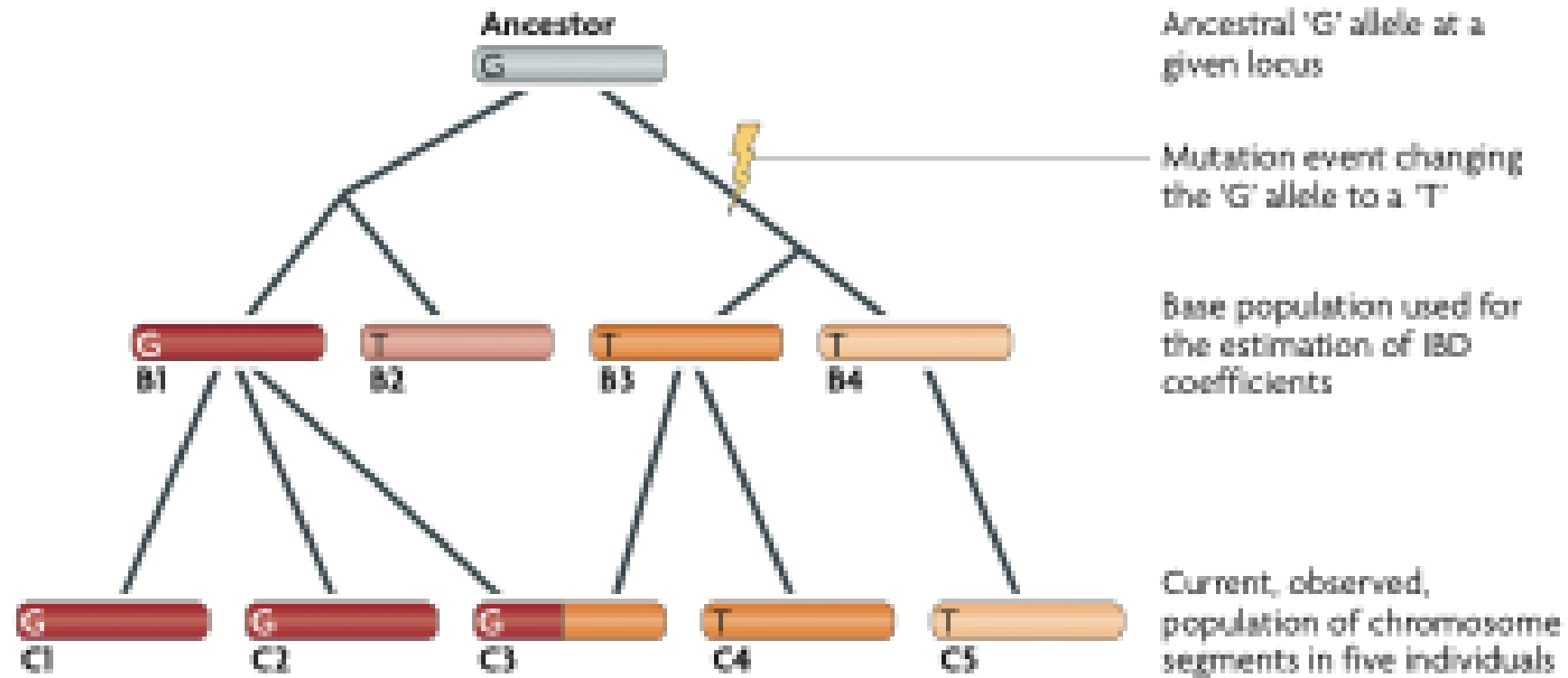


*shared
ancestor*

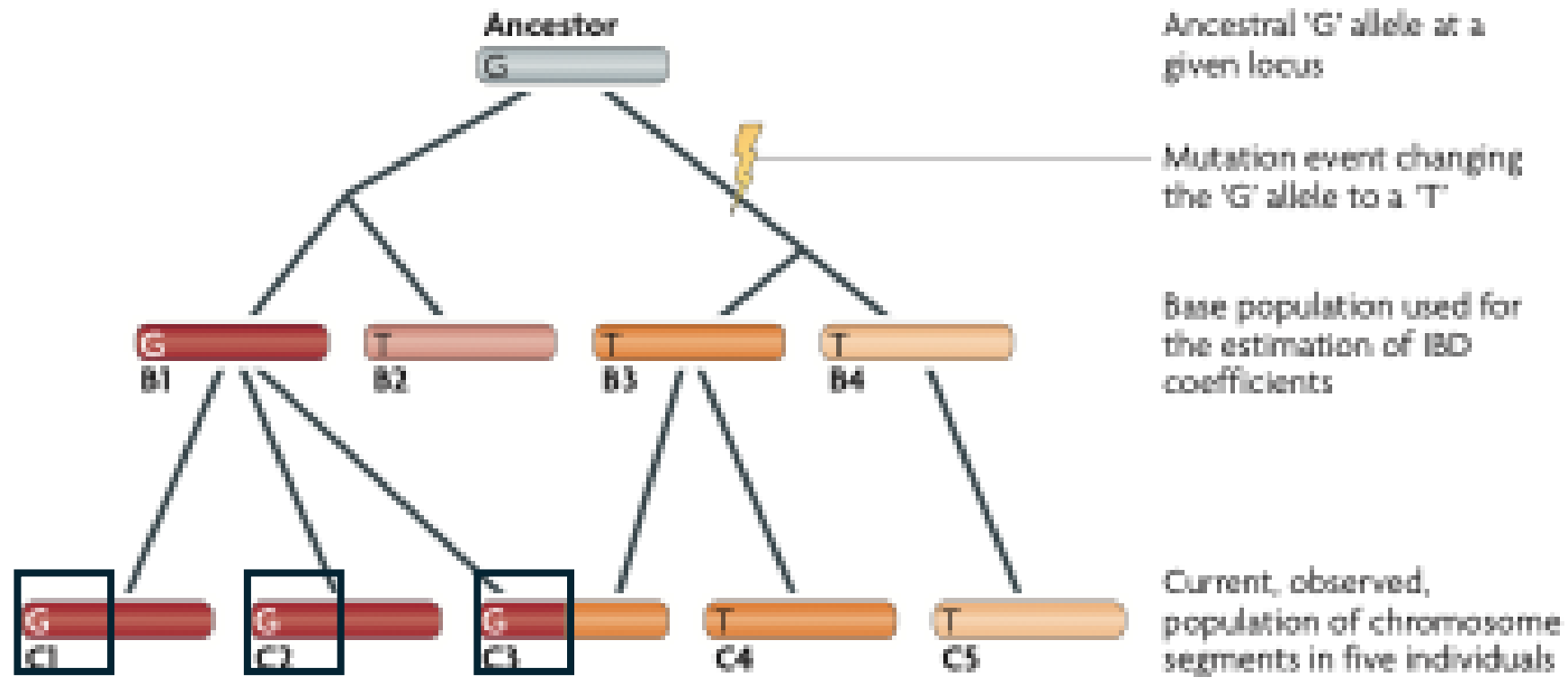
*Same alleles AND inherited from
a common ancestor*

Two alleles are IBS if they carry the same nucleotide—regardless of ancestry. IBS includes IBD, but not vice versa.

Specific example



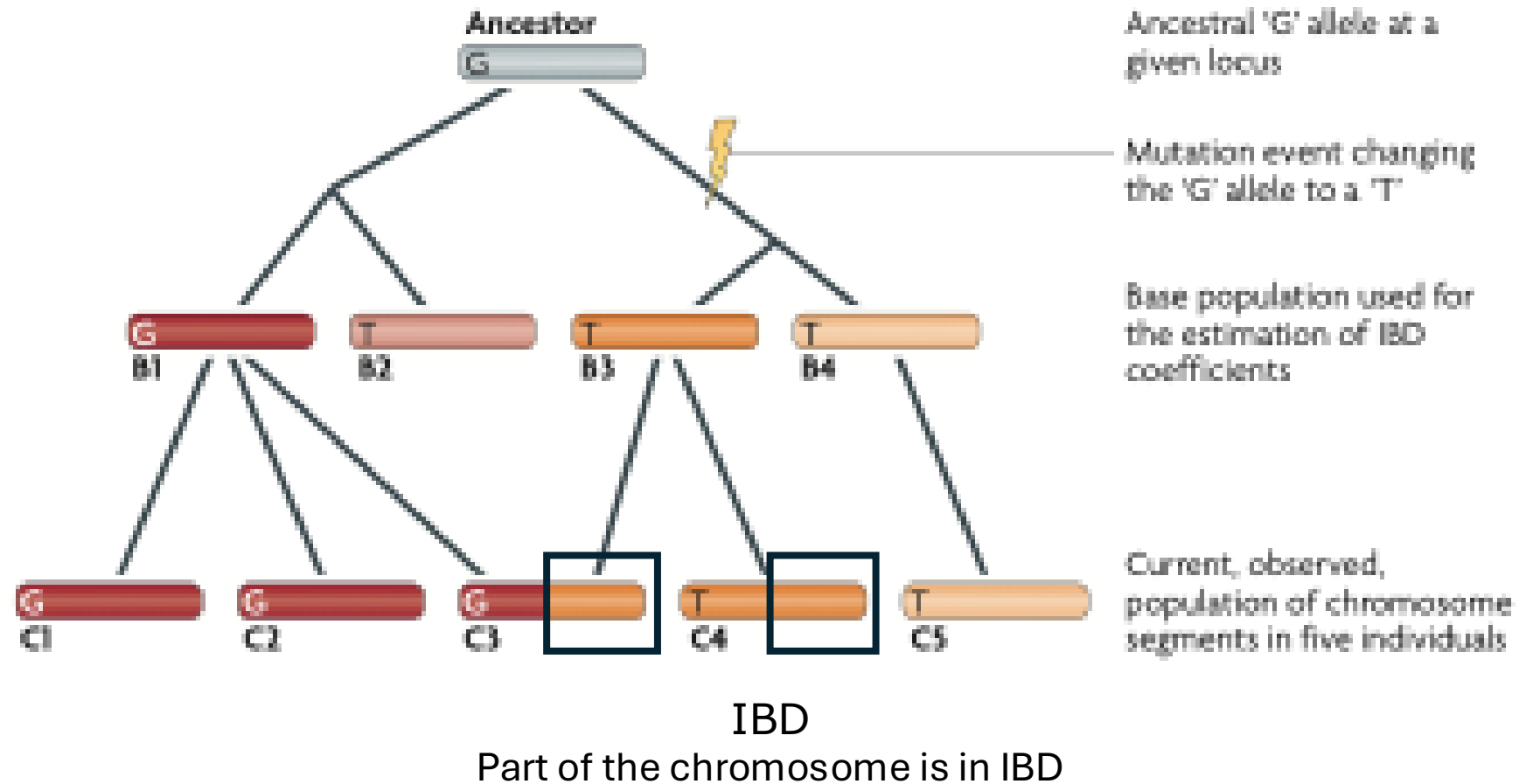
Specific example



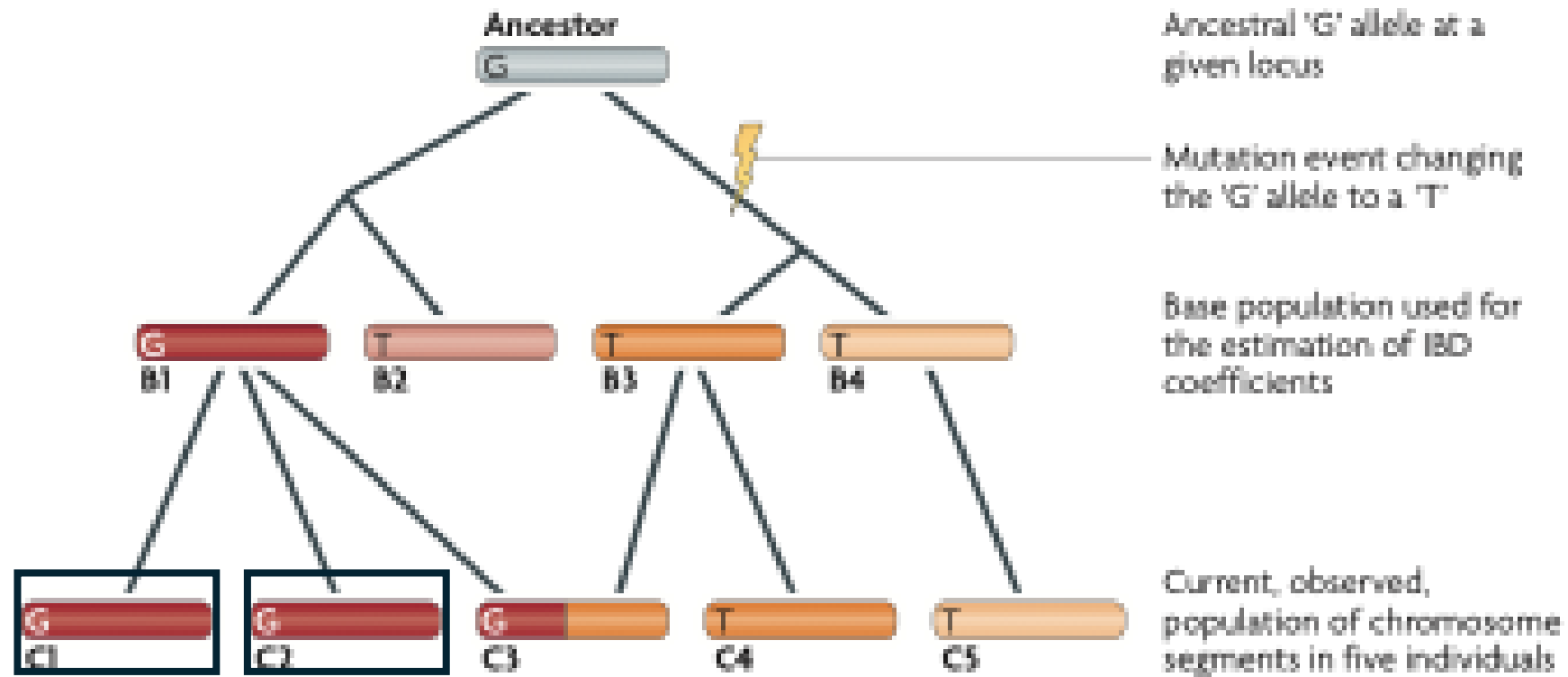
IBD

The **G** alleles in **C1**, **C2** and **C3** are IBD to each other as all three descend from the G allele in B1

Specific example



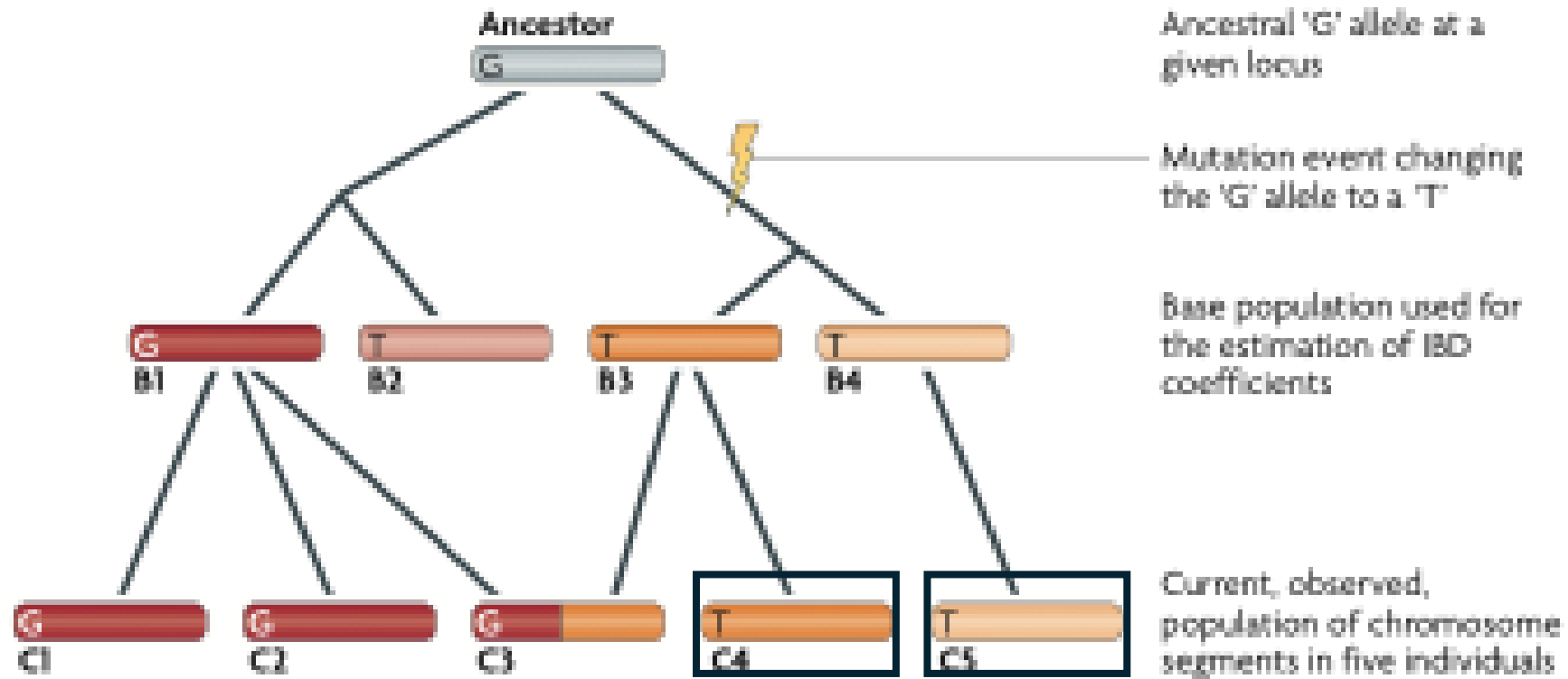
Specific example



IBD

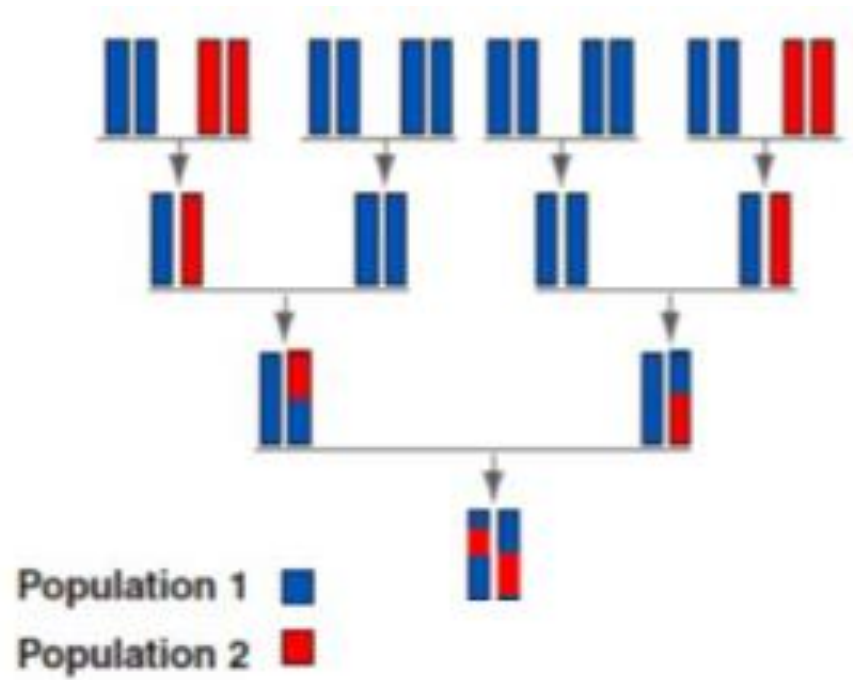
The **whole chromosome** in **C1** and **C2** are IBD because they descend from B1 without recombination

Specific example

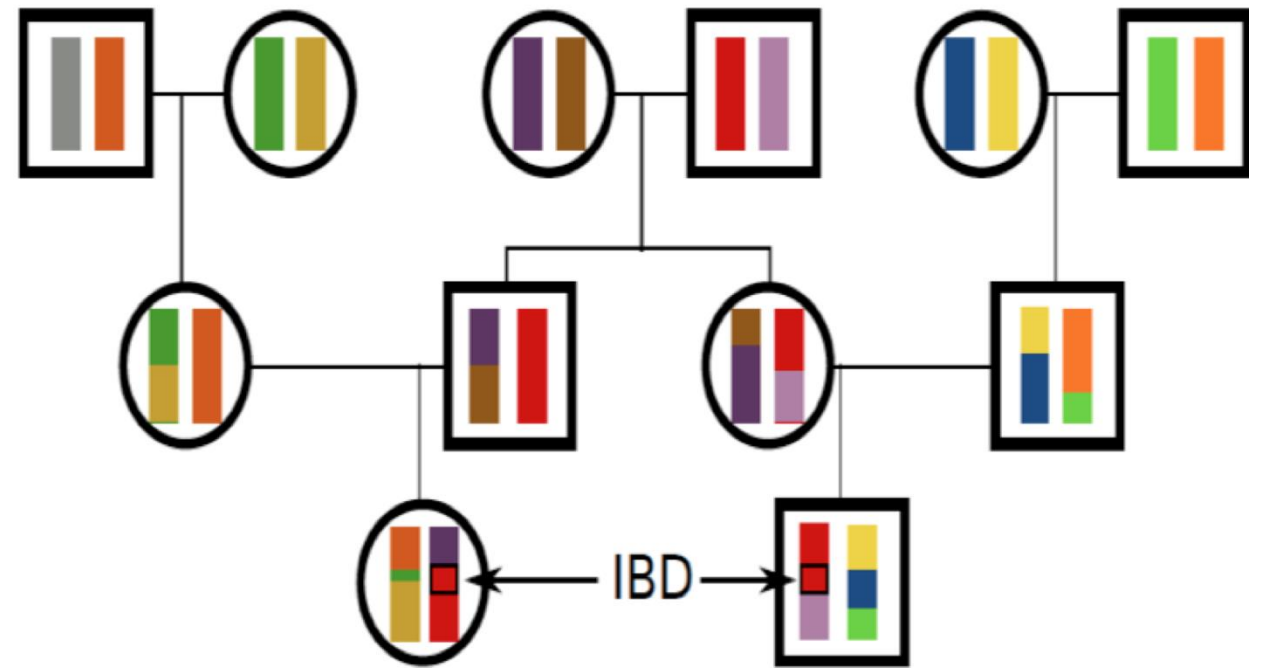
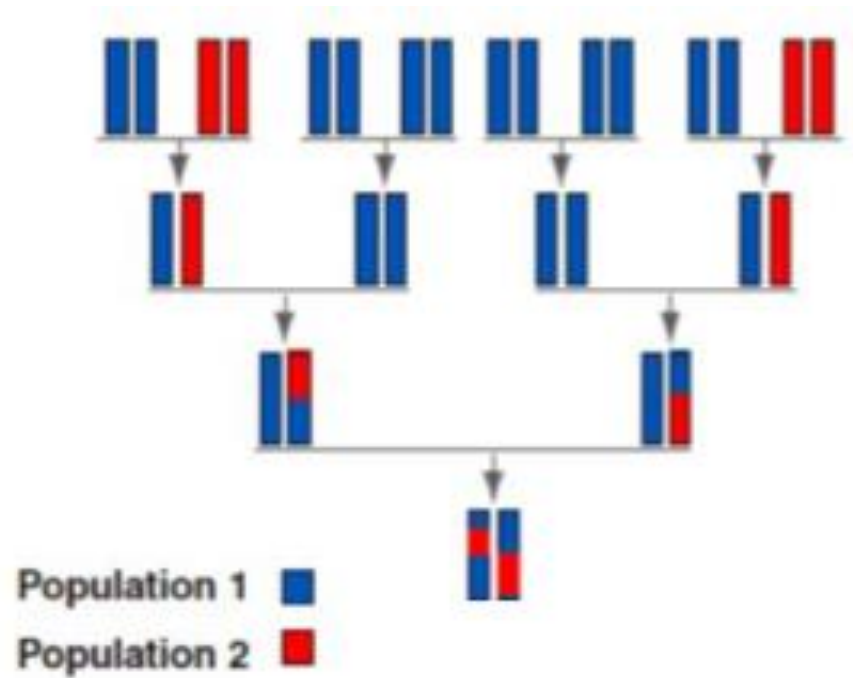


IBS not IBD
As the chromosomes descend from **different** ancestors

Pedigree



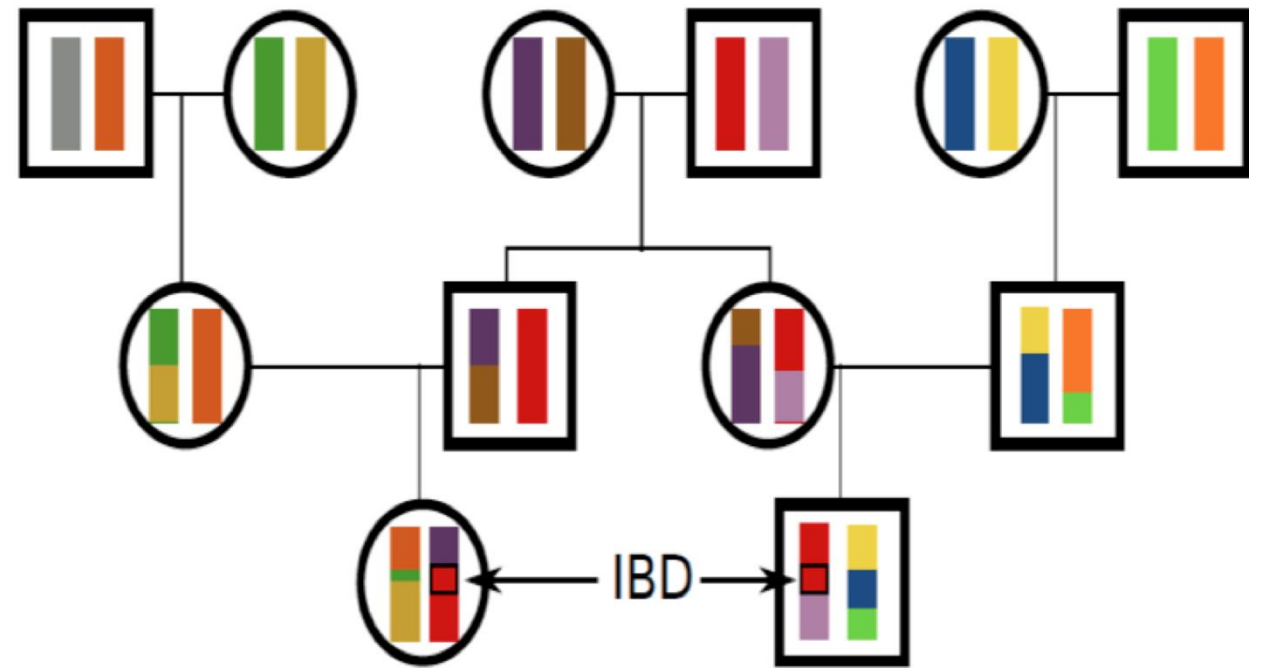
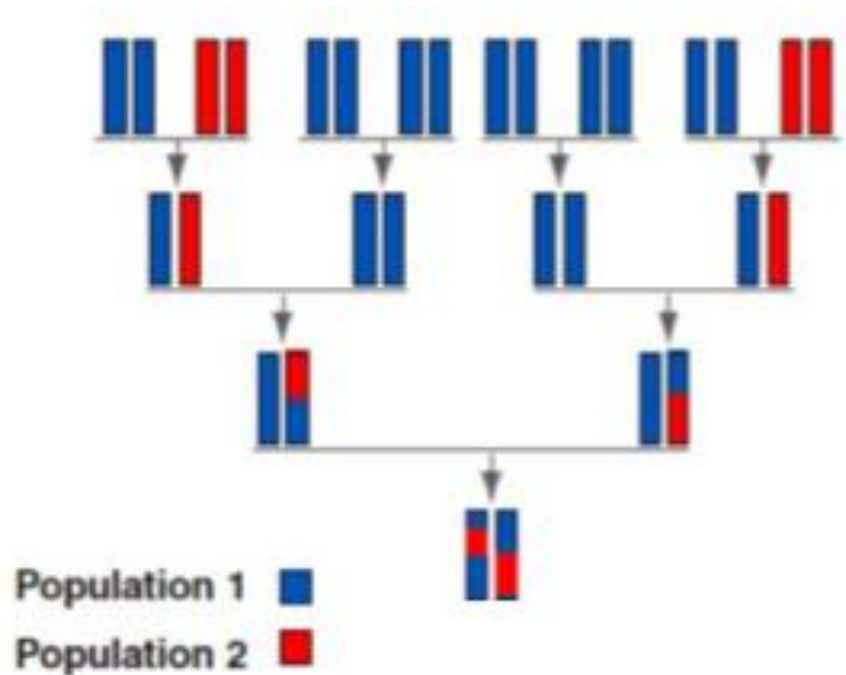
IBD - Identity by Descent



IBD - Identity by Descent

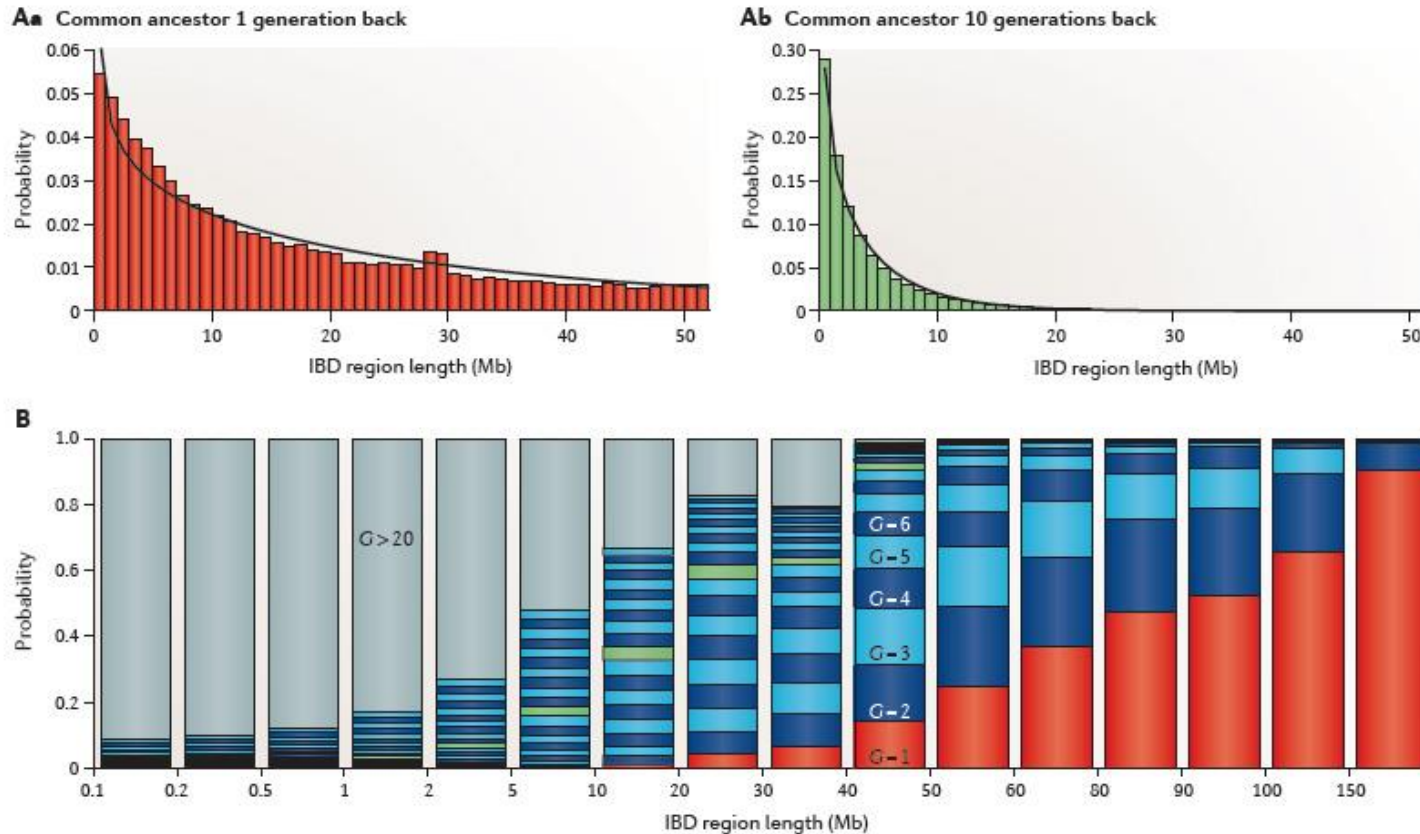
We will define two alleles to be identical by descent (IBD) if they are identical due to transmission from a common ancestor in the past few generations.

The IBD is the variability that cannot be explained by the population frequency



Calculating IBD

What lengths do IBD segments have for a known ancestor age?



Given an IBD segment of this length, how probable is each ancestor generation?

A recent common ancestor tends to produce **long IBD segments**, whereas an older common ancestor produces **shorter segments**.

How many generations back should I pick?

The shortest IBD segment length you can call reliably.

We stick with **recent shared ancestry**, because with deeper times it gets increasingly hard to distinguish from IBS.

IBD States: IBD0, IBD1, IBD2

For any pair of diploid individuals, at each locus they share 0, 1, or 2 alleles IBD

IBD0

No alleles shared IBD



Unrelated

IBD1

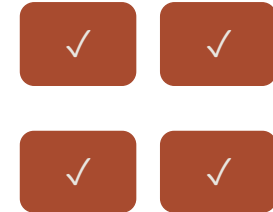
One allele shared IBD



Parent-offspring

IBD2

Both alleles shared IBD



MZ twins / inbred

IBD and relatedness coefficients

How much a pair shares IBD is quantified in several ways :

$R = (k_0, k_1, k_2)$: fractions of genome with 0, 1 and 2 alleles IBD (**IBD states distribution**)

k_0 – probability of sharing **no** IBD alleles at a locus

k_1 – probability of sharing **one** IBD alleles

k_2 – probability of sharing **two** IBD alleles

$$k_0 + k_1 + k_2 = 1$$

r (**coefficient of relatedness**): expected fraction of alleles of shared IBD

$$r = \frac{1}{2}k_1 + k_2 \Rightarrow r = 0 \times k_0 + \frac{1}{2}k_1 + 1 \times k_2$$

ϕ (kinship coefficient): probability that 2 random alleles are IBD ($\phi = 0.5r$)

$$\phi = \frac{1}{4}k_1 + \frac{1}{2}k_2$$

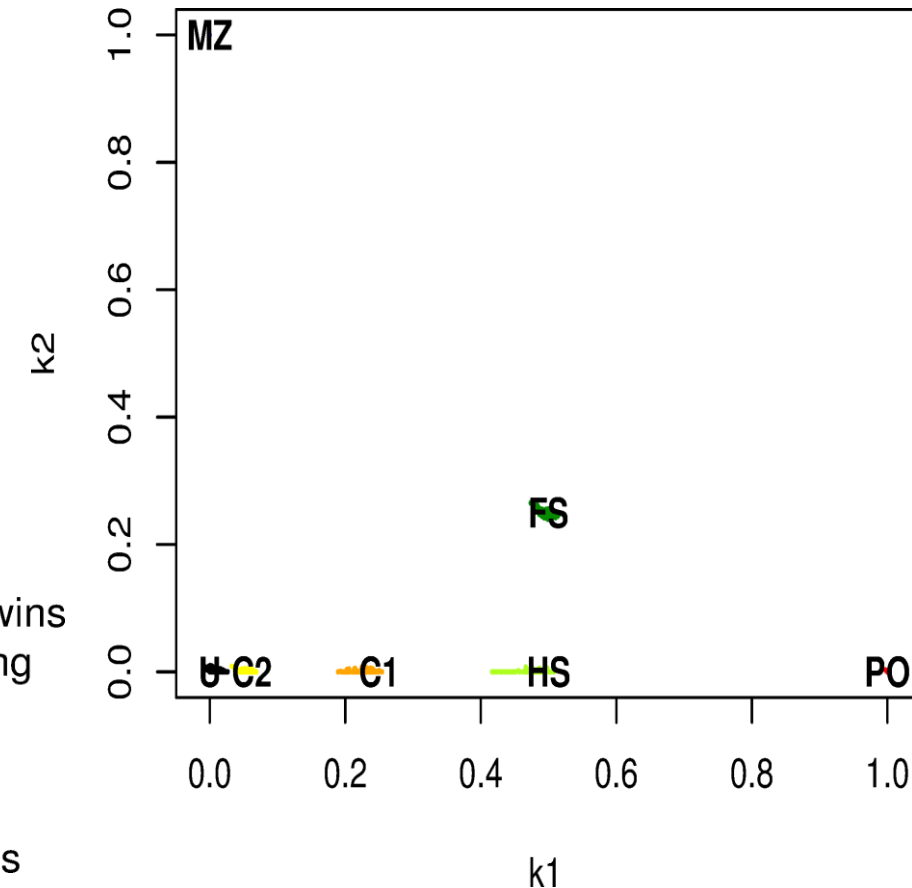
Inference of relationships

We expect different relationships to have different values of R , e.g.:

Relationship	k_0	k_1	k_2
Monozygotic twins	0	0	1
Parent-offspring	0	1	0
Siblings	0.25	0.5	0.25
Halfsiblings/Uncle-nephew/grandparent-child	0.5	0.5	0
First cousins	0.75	0.25	0
Second cousins	0.9375	0.0625	0
Unrelated	1	0	0

Hence we can (often) use R to infer how two individuals are related

- Unrelated
- Monozygotic twins
- Parent-offspring
- Full siblings
- Half siblings
- Cousins
- Second cousins



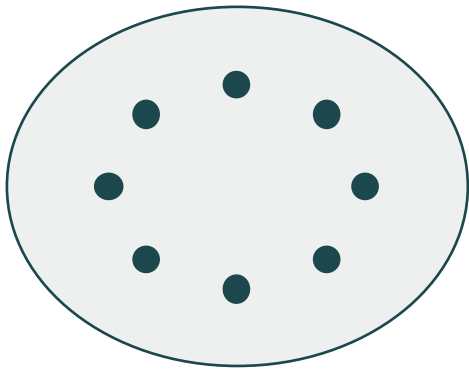
All together

Relationship	k_0	k_1	k_2	r	ϕ
Identical twins / Self	0	0	1	1	0.5
Parent-offspring	0	1	0	0.5	0.25
Full siblings	1/4	1/2	1/4	0.5	0.25
Half siblings	1/2	1/2	0	0.25	0.125
First cousins				0.125	0.0625
Unrelated	1	0	0	0	0

$$(k_0, k_1, k_2) \rightarrow \phi = \frac{1}{4}k_1 + \frac{1}{2}k_2 \rightarrow r = 2\phi = \frac{1}{2}k_1 + k_2$$

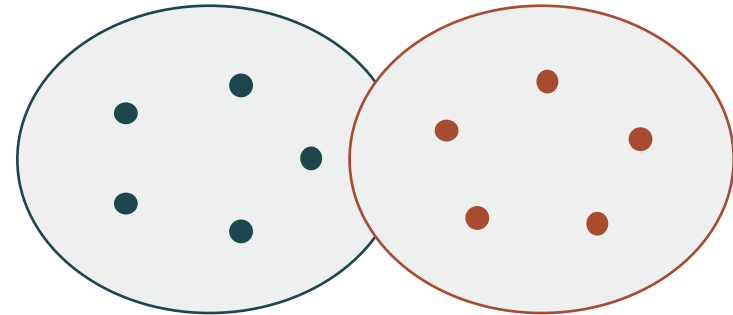
Population structure confounds Relatedness

Scenario A: Single Population



*Shared alleles reflect
IBD (recent common ancestry)*

Scenario B: Structured Populations

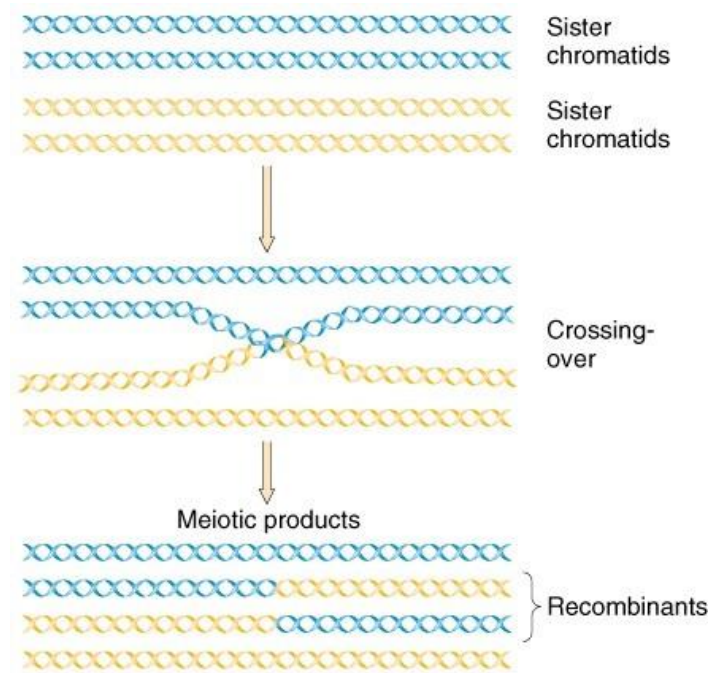
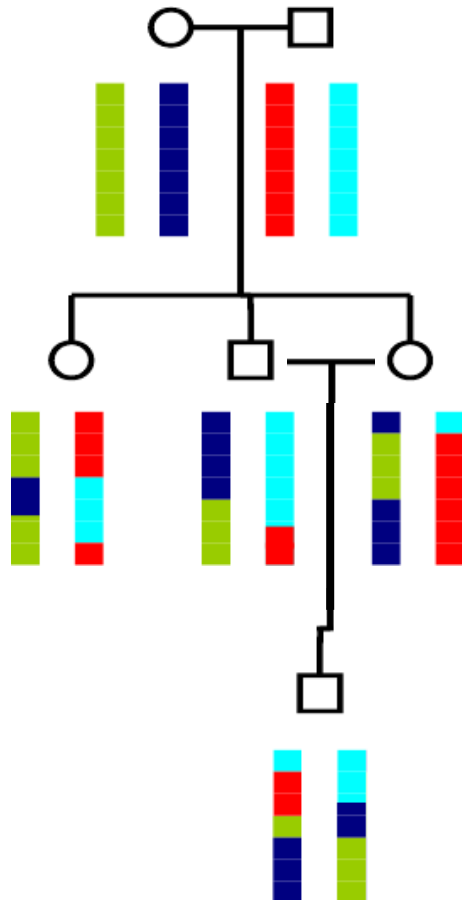


*Shared alleles reflect
population ancestry, not IBD!*

Problem: Traditional relatedness estimators use population allele frequencies. When structure exists, individuals from the same sub-population appear "related" even when they are not.

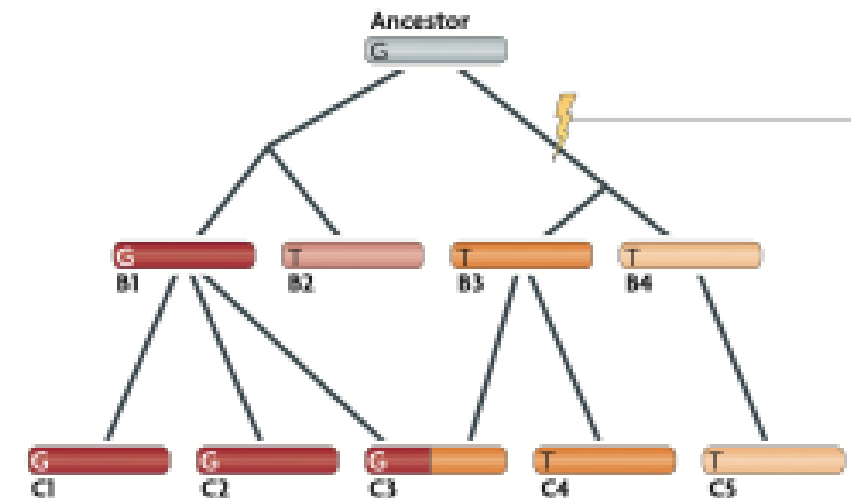
A definition for IBD

Two alleles are *Identical by descent* (IBD) if they are direct copies of a single ancestral allele



Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci

Population structure matters for IBD calling



Relatedness is important

1

Analysis

Many analysis assume that all individuals analyzed are unrelated and if related individuals are not removed or controlled for they can lead to highly inflated false-positive rates.

2

Diseases

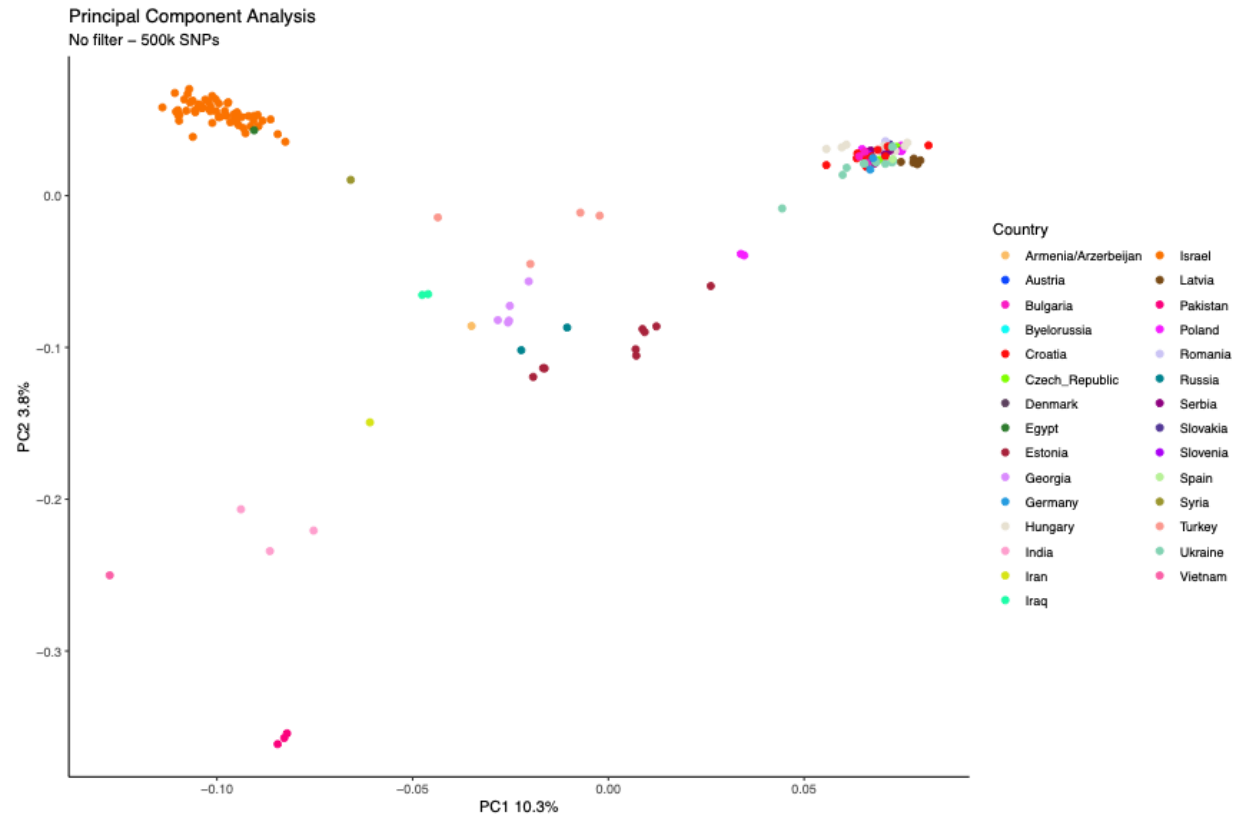
3

Inbreeding

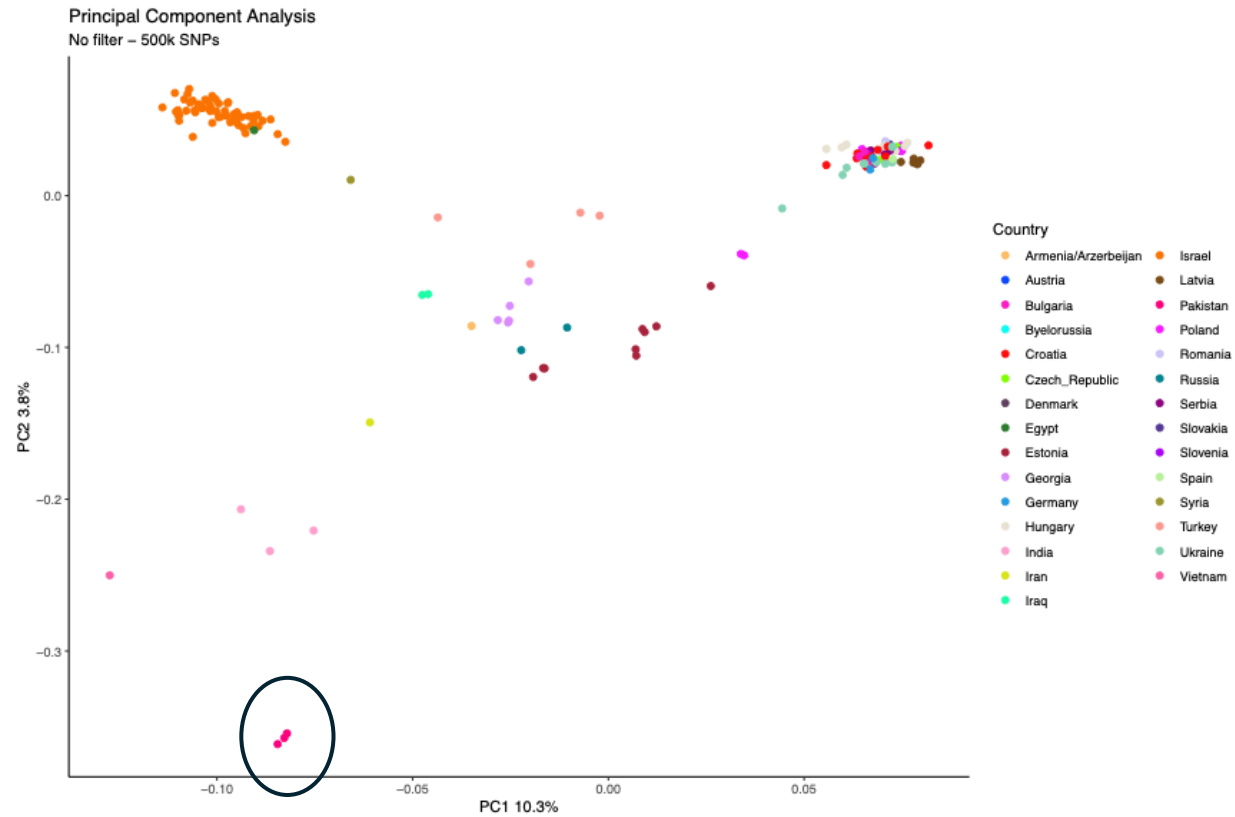
4

Conservation

PCA results – population structure

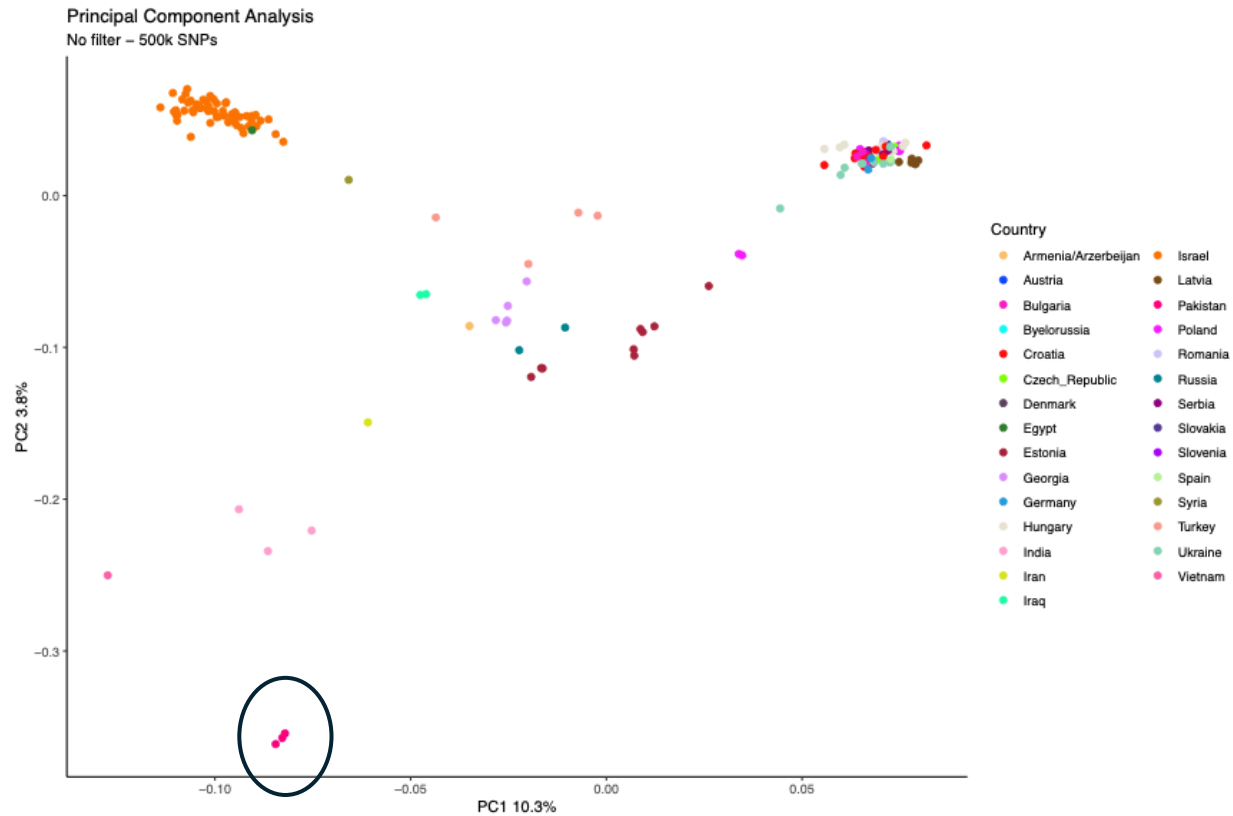


PCA results – population structure



Pakistan

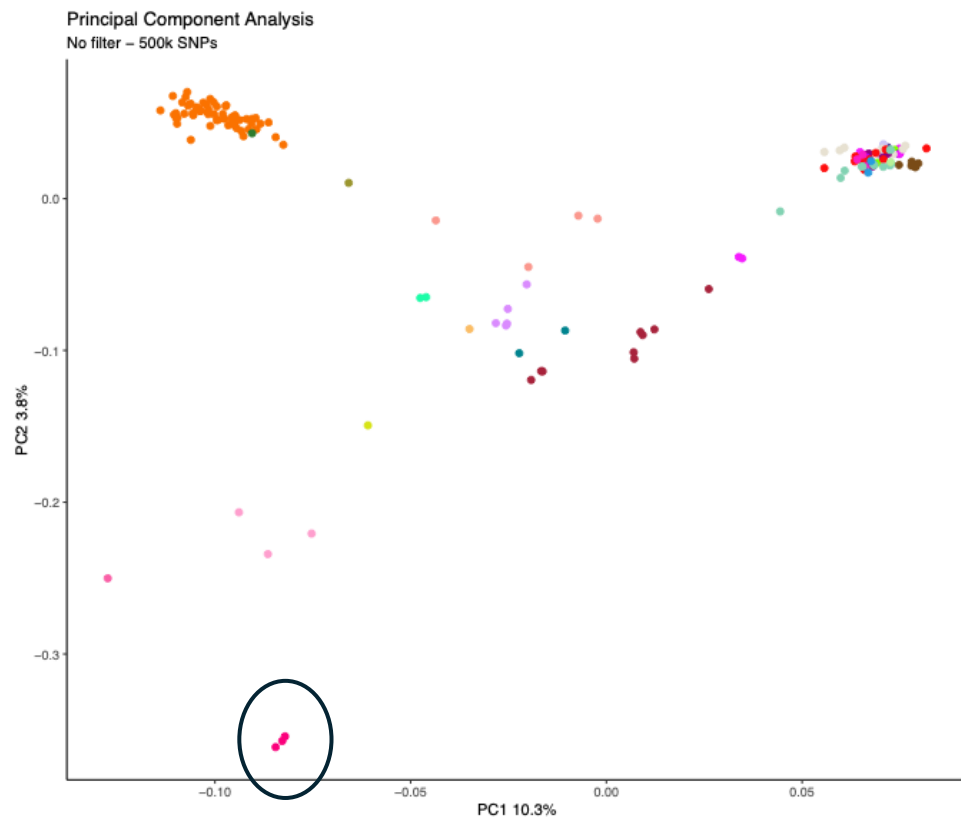
PCA results – population structure



Pakistan

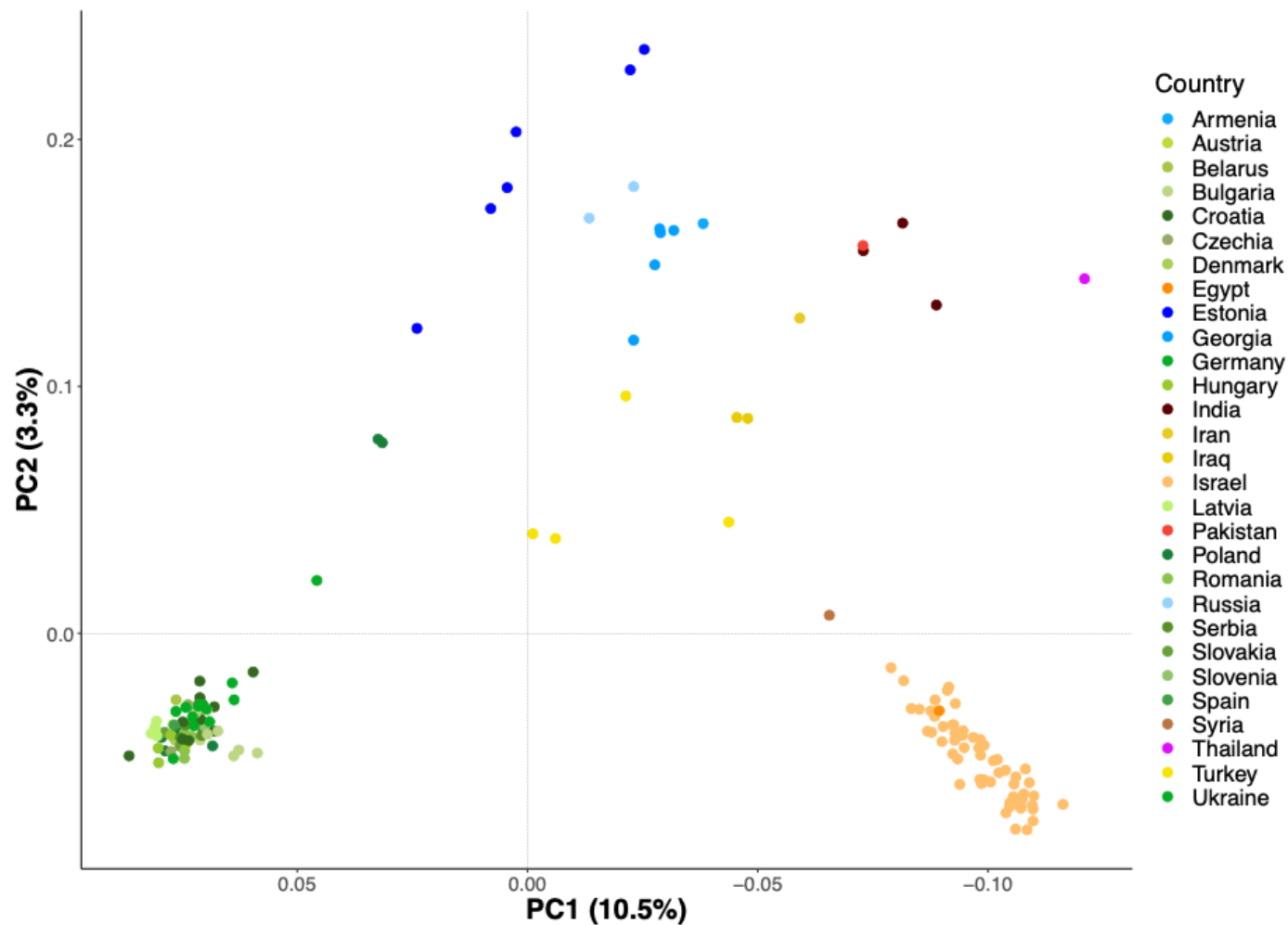
Kinship 0.48

PCA results – population structure

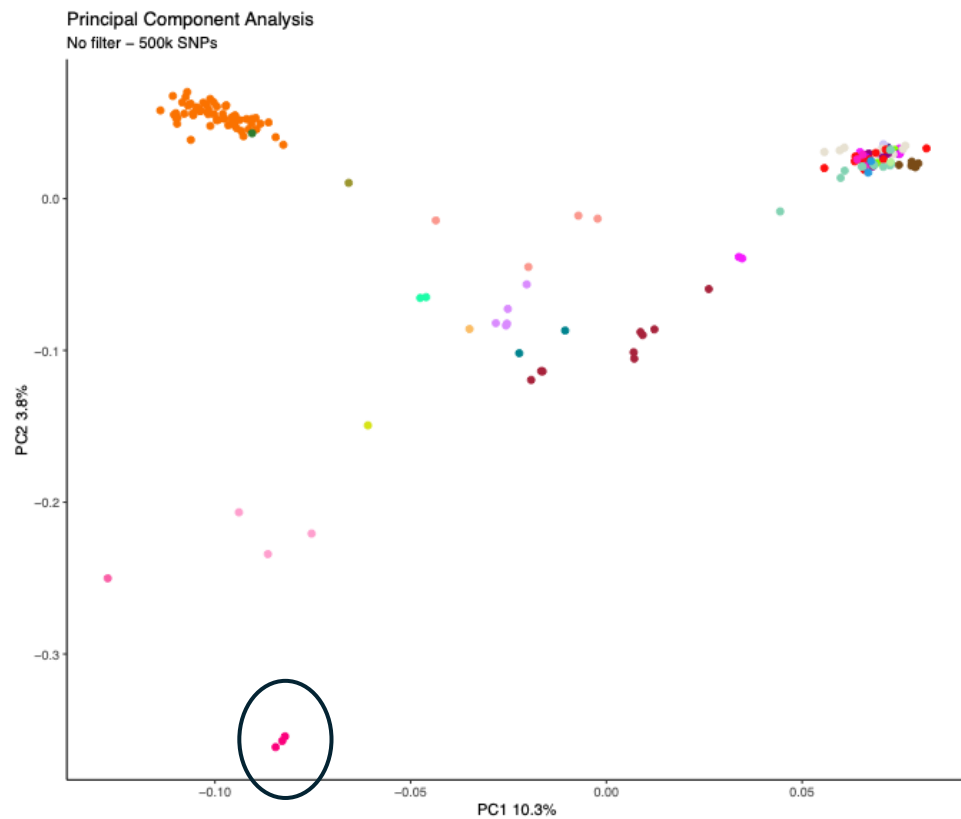


Pakistan

Kinship 0.48

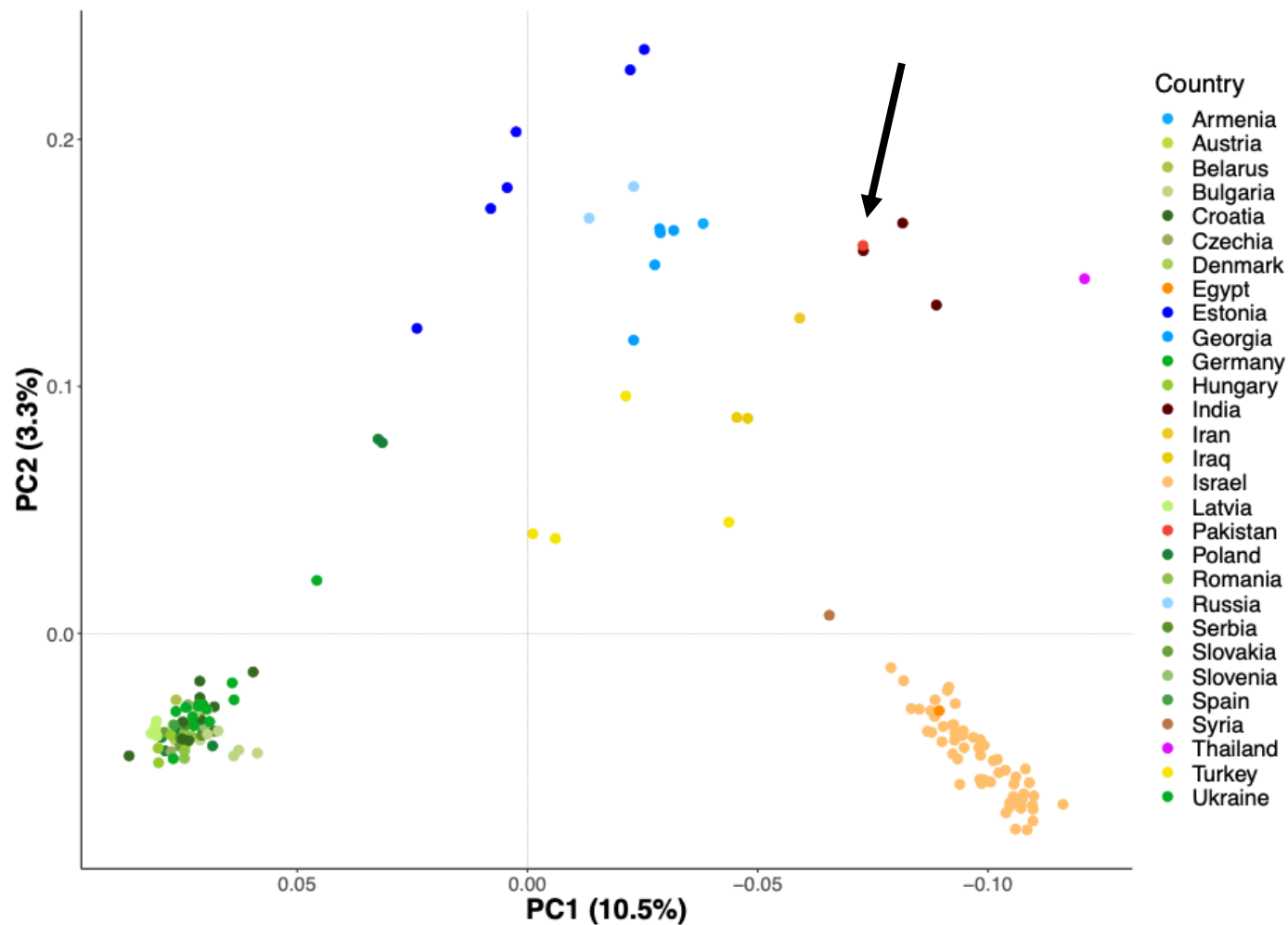


PCA results – population structure



Pakistan

Kinship 0.48



Relatedness is important

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PCA results - population structure

Many analysis assume that all individuals analyzed are unrelated and if related individuals are not removed or controlled for they can lead to highly inflated false-positive rates.

2

Diseases

If affected relatives share a chromosomal segment **IBD** more often than expected by chance, that segment may contain a disease-causing variant.

3

Inbreeding

4

Conservation

Relatedness is important

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Diseases

If affected relatives share a chromosomal segment **IBD** more often than expected by chance, that segment may contain a disease-causing variant.

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Inbreeding

IBD is the mechanism behind inbreeding: an individual is inbred at a locus when its two homologous alleles are **identical by descent** from the same ancestral allele.

High relatedness produces long runs of homozygosity (**ROH**)

4

Conservation

Relatedness is important

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Conservation

A pedigree-based management strategy that minimizes mean kinship is effective at **mitigating drift** and **inbreeding**, while preserving genetic diversity and evolutionary potential to curtail extinction risk

How to calculate it? Relatedness

King

relateAdmix

Li & Ro

KING – Kinship-based inference

KING-robust kinship estimator (Manichaikul et al. 2010)

$$\hat{\varphi}_{ij} = \frac{[IBS2^* - 2 \cdot IBS0]}{[4 \cdot \min(HET_i, HET_j)]}$$

IBS0 = SNPs where both are homozygous but for different alleles

IBS2* = SNPs where both are heterozygous

HET = proportion of heterozygous sites per individual

Expected kinship thresholds

$$\varphi > 0.354$$

Duplicate / MZ twin

$$0.177 - 0.354$$

1st degree (parent/offspring, full sib)

$$0.088 - 0.177$$

2nd degree (half-sib, grandparent, uncle)

$$0.044 - 0.088$$

3rd degree (1st cousins)

$$\varphi < 0.044$$

Unrelated

Why KING is robust to structure

Most earlier estimators (e.g., PLINK --genome) assume random mating and break down across diverged populations. KING-robust is approximately unbiased even for between-population pairs, because IBS0 acts as a correction for allele frequency divergence.

Limitation: still slightly biased with strong population structure. For structured datasets → use relateAdmix or PC-Relate.

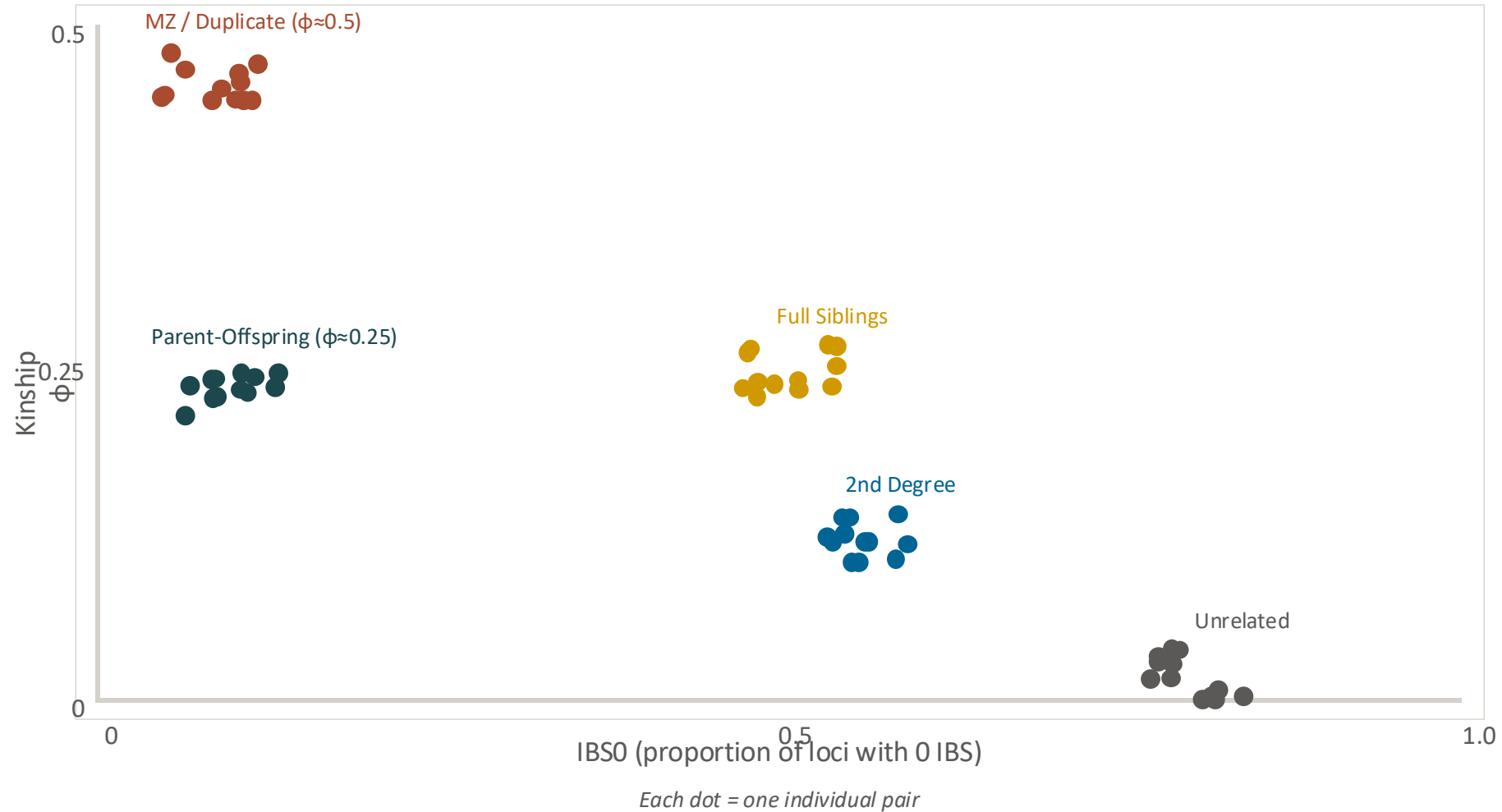
KING in PLINK 2

```
# PLINK 2 KING-robust
plink2 --bfile dataset \
  --make-king-table \
  --king-table-filter 0.044 \
  --out kinship_results

# Columns: ID1 ID2 NSNP HETHET IBS0 KINSHIP
```

Relationship Categories: Kinship vs IBS0

KING-robust classifies pairs using kinship coefficient and IBS0 proportion



relateAdmix: Why ancestry matters

The Problem

Standard estimators (PLINK, even KING) use average population allele frequencies as background. **Admixed individuals draw alleles from MULTIPLE ancestral populations**—so the "background frequency" varies along the genome.

Using a single-population baseline inflates false kinship between admixed individuals from the same source populations.

Example: Two African Americans with 80% West African ancestry share many alleles due to shared ancestry, inflating apparent relatedness under standard estimators.

KING handles global structure but can still be biased in admixed samples.

relateAdmix Solution

Decomposes allele sharing into:

1. Sharing due to IBD (recent common ancestry)
2. Sharing due to population ancestry

Uses **local ancestry** inference to estimate how much sharing is expected from ancestry alone, then subtracts this to get the IBD component.

Key advantage: Works in admixed populations where KING and Li/Ro are biased.

Li & Ro — IBD Segment Statistics

Beyond point estimates: IBD sharing

Instead of estimating a single kinship coefficient per pair, Li & Ro statistics characterise the distribution of IBD-shared segments across the genome. This captures not just HOW MUCH is shared, but WHERE and in WHAT SIZES.

Key statistics

Li

Total length of genome shared IBD (Morgans or Mb). Sensitive to recent relatedness.

Ro

Proportion of genome in IBD segments > threshold length. More specific to close relatives.

N_seg

Number of IBD segments. Long segments = recent ancestry; many short = old ancestry.

L_seg

Average segment length. Shorter segments → older shared ancestor (more recombination time).

Conceptual IBD segment diagram



IBD segments (shared regions, same colour)

Longer segments → more recent common ancestor
Many short segments → ancient relationship

Time for exercises

Please do the Relatedness and stop before the
Fst in Wildebeest



Key takeaways

1

Relatedness

IBD is the biological foundation. All relatedness estimators approximate the kinship coefficient ϕ —the **probability that two alleles are inherited from a common ancestor**.

Related individuals distort all downstream statistics leading to wrong conclusions

KING-robust is the recommended default pairwise estimator. Use relateAdmix for admixed populations, ngsRelate for low-coverage data.

2

Fst

Fst quantifies **population differentiation**. It can be defined as the ratio of the variance in allele frequencies among populations to the total genetic variance.

Low-diversity populations can show inflated FST even with minor allele frequency differences.

Population structure can inflate the Fst estimation

- Manichaikul et al. 2010 (KING paper)
- Moltke & Albrechtsen 2014 (relateAdmix paper)
- Korneliussen et al. 2014 (ANGSD paper)
- Weir & Cockerham 1984 (FST estimator—the canonical reference)